

生态研究前沿速览_2026-04-08

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【土壤与生物地球化学】

1. Characterisation and quantification of organic carbon burial using a multiproxy approach in saltmarshes from Aotearoa New Zealand

☐ Biogeosciences | Mon, 30 Mar 2026 08:

☐ Abstract:

Characterisation and quantification of organic carbon burial using a multiproxy approach in saltmarshes from Aotearoa New Zealand Olga Albot, Joshua Ratcliffe, Richard Levy, Sebastian Naeher, Daniel J. King, Catherine Ginnane, Jocelyn Turnbull, Mary Jill Ira Banta, Christopher Wood, Jenny Dahl, Jannine Cooper, and Andy Phillips Biogeosciences, 23, 2155–2178, <https://doi.org/10.5194/bg-23-2155-2026>, 2026 Saltmarshes store carbon in their soils, contributing to climate change mitigation. We analysed five sites across New Zealand and found that carbon storage and accumulation rates vary widely with geomorphic setting, land use history and sediment inputs. Plant material was a major source of carbon in the soil and has been preserved in basal sediments for several centuries. These results impr

☐ <https://doi.org/10.5194/bg-23-2155-2026>

2. Emerging Climate Signals in Tropical Oxygen Minimum Zones

☐ Biogeosciences | Tue, 07 Apr 2026 08:

□ Abstract:

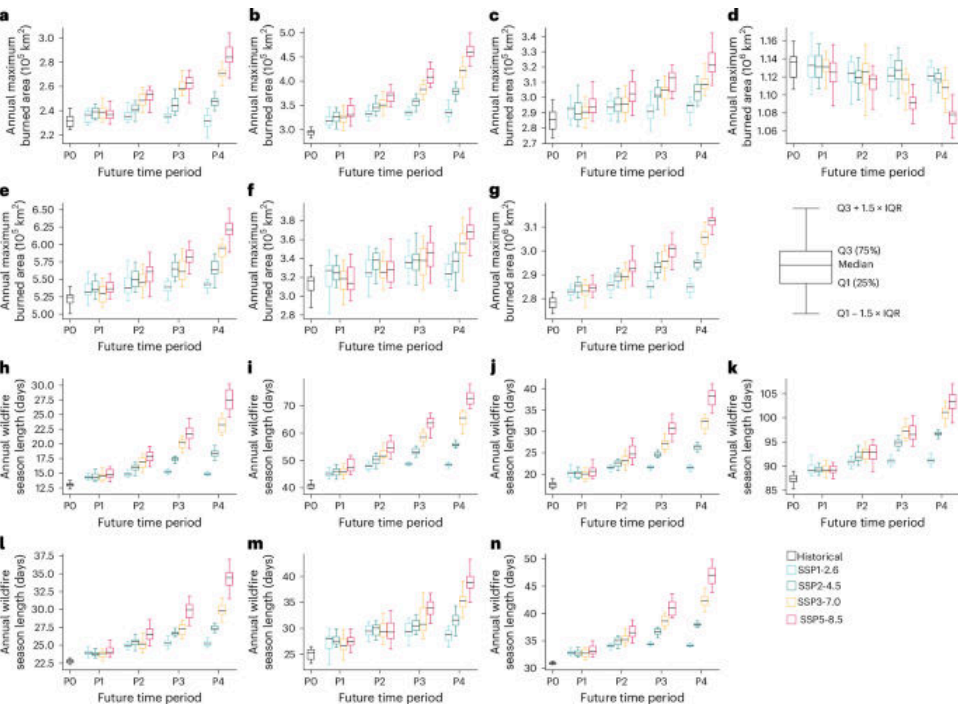
Emerging Climate Signals in Tropical Oxygen Minimum Zones Mathieu Delteil, Marina Lévy, and Laurent Bopp Biogeosciences, 23, 2205–2233, <https://doi.org/10.5194/bg-23-2205-2026>, 2026 The ocean is losing oxygen due to climate change, threatening ecosystems, especially in naturally low-oxygen areas called Oxygen Minimum Zones (OMZs). Using the IPSL-CM6A-LR Large Ensemble, this study identifies when climate-driven changes in OMZ volumes and regional deoxygenation emerge from natural variability. We highlight hemispheric asymmetries due to ocean ventilation and provide model-based estimates for the timing of detectable OMZ evolution.

□ <https://doi.org/10.5194/bg-23-2205-2026>

【环境与气候】

3. Wildfire risk for species under climate change

□ Nature Climate Change | 2026-04-06



□ Abstract:

Nature Climate Change, Published online: 06 April 2026; doi:10.1038/s41558-026-02600-5The authors consider risks to global biodiversity from wildfire under climate change. They show increased risk to 83.9% of species pre-identified as wildfire vulnerable, with high risks for species with small ranges, high conservation concern and those in South America, Australia and South Asia.

□ <https://doi.org/10.1038/s41558-026-02600-5>

4. Distributional consequences of climate policy

□ Nature Climate Change | 2026-04-03



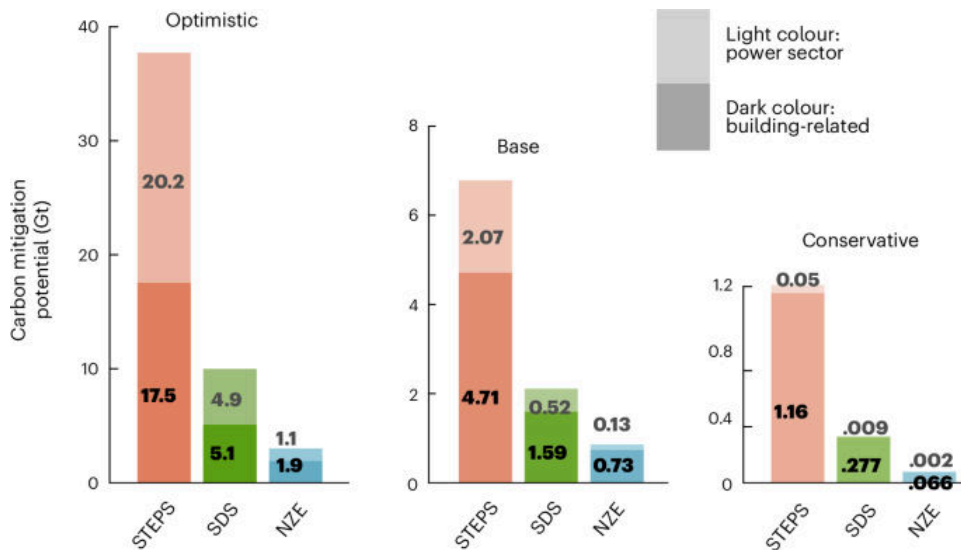
☐ Abstract:

Nature Climate Change, Published online: 03 April 2026; doi:10.1038/s41558-026-02612-1 Carbon pricing can be a cost-effective way to cut carbon dioxide emissions, but only if it is politically sustainable. Two recent papers document how carbon pricing can create winners and losers, while also showing how these shortcomings can be addressed by careful policy design.

☐ <https://doi.org/10.1038/s41558-026-02612-1>

5. Global energy and climate benefits from photovoltaics integrated in building façades

☐ Nature Climate Change | 2026-04-03



☐ Abstract:

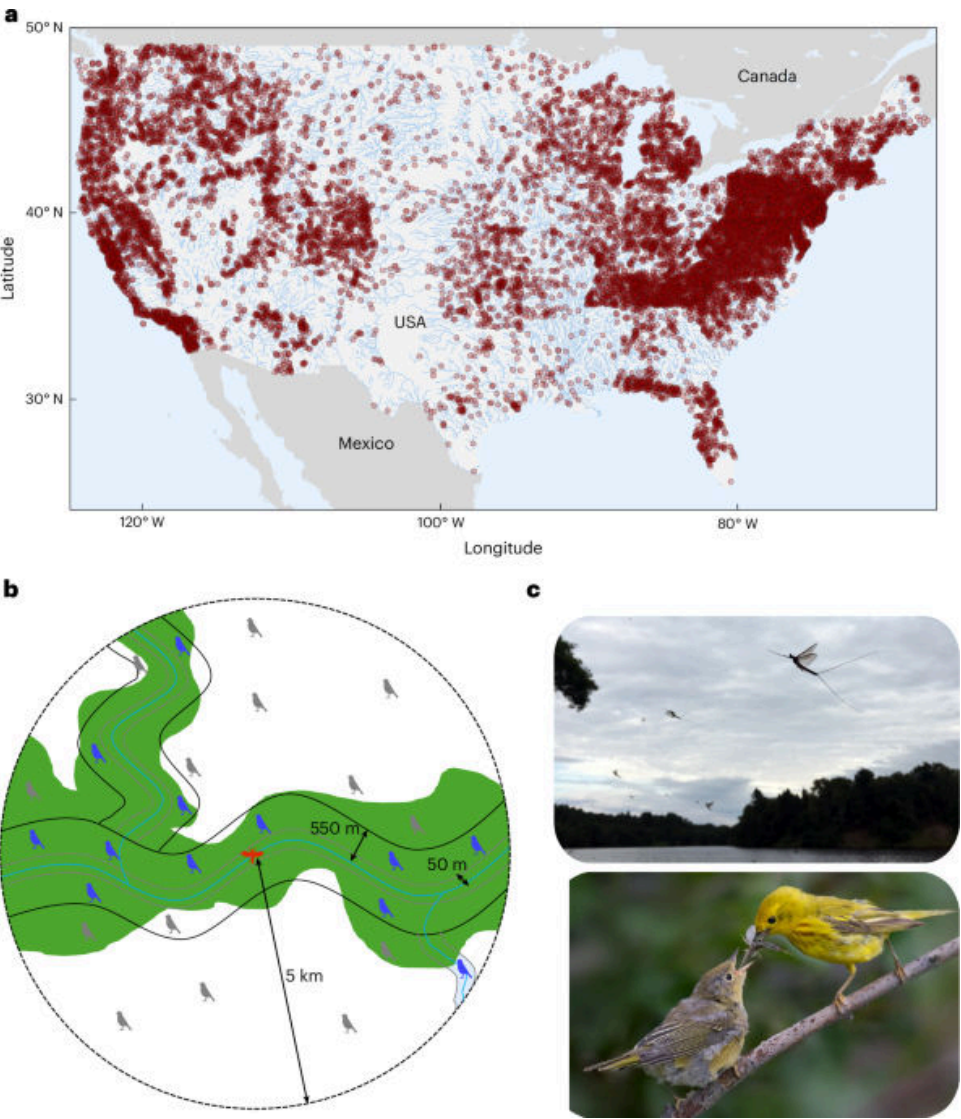
Nature Climate Change, Published online: 03 April 2026; doi:10.1038/s41558-026-02621-0Global modelling shows that integrating photovoltaics in the façades of buildings could deliver substantial electricity generation, building energy savings and emissions reductions —and highlights an underexplored opportunity for urban energy transition and climate mitigation.

☐ <https://doi.org/10.1038/s41558-026-02621-0>

【生态学核心】

6. Cross-ecosystem linkages between freshwater insects and riparian birds across the USA

☐ Nature Ecology & Evolution | 2026-04-07



☐ Abstract:

Nature Ecology & Evolution, Published online: 07 April 2026; doi:10.1038/s41559-026-03041-1Aquatic insects such as mayflies, stoneflies and caddisflies are widely used for freshwater biomonitoring and could provide resource

pulses to terrestrial insectivores. Here the authors show a positive relationship between the species richness of these insects and the prevalence of riparian birds across the conterminous USA.

☐ <https://doi.org/10.1038/s41559-026-03041-1>

【综合顶刊】

7. Massive wildfires followed oceanic anoxic events during the Late Devonian Frasnian-Famennian mass extinction

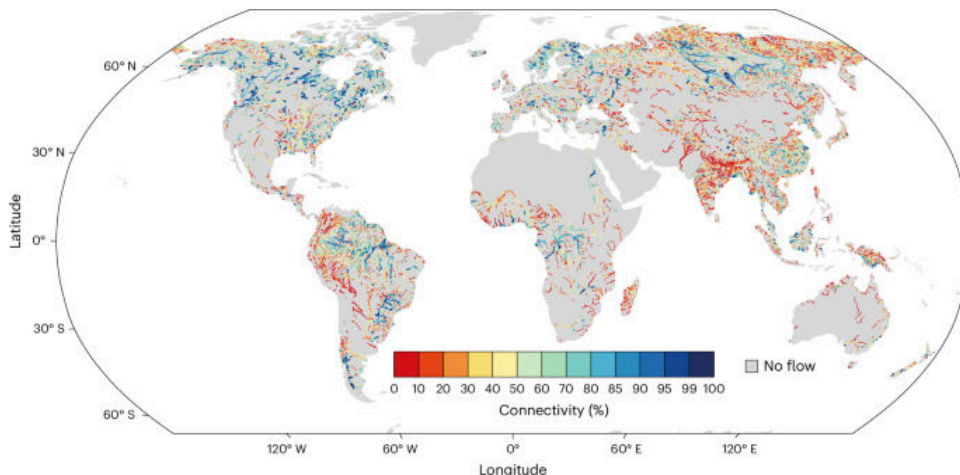
☐ Science Advances | 2026-04-01T07:00:00Z

☐ <https://doi.org/10.1126/sciadv.ady4534>

【环境与气候】

8. Global net increase in surface water connectivity in river–floodplain systems

☐ Nature Geoscience | 2026-04-02



☐ Abstract:

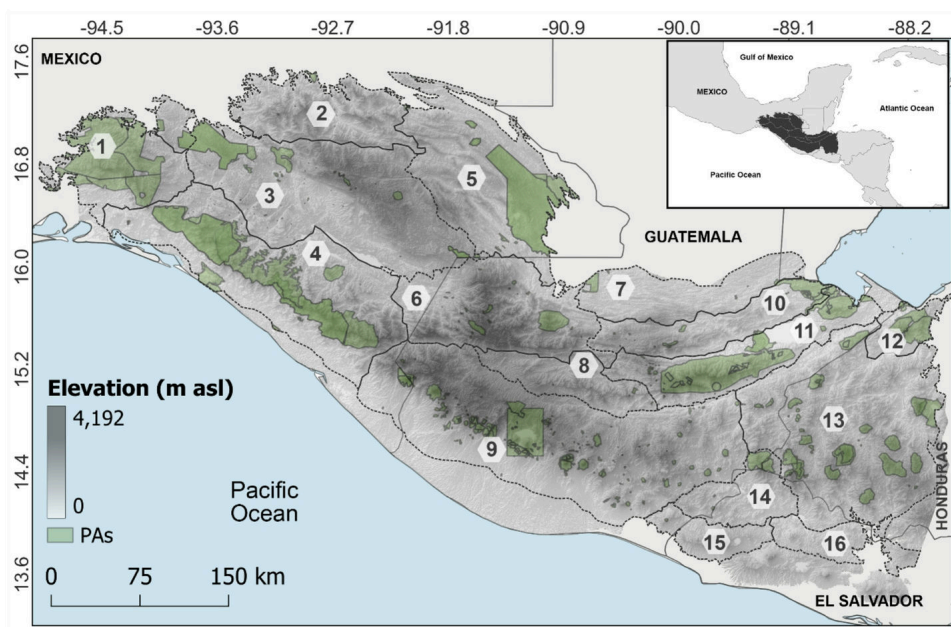
Nature Geoscience, Published online: 02 April 2026; doi:10.1038/s41561-026-01953-y The flow of water between rivers and floodplains—surface water connectivity—showed a net global increase from 1984 to 2019 driven by climate and anthropogenic changes and shaping sediment transport, according to a study of satellite observations.

☐ <https://doi.org/10.1038/s41561-026-01953-y>

【保护与生物多样性】

9. Climate-driven erosion of richness and evolutionary diversity and conservation shortfall in a salamander hotspot

☐ Biodiversity and Conservation | 2026-04-07



☐ Abstract:

Anthropogenic climate change poses a significant threat to biodiversity, particularly for amphibians inhabiting tropical mountain systems. Due to their reliance on humid montane forests and moderate thermal conditions, plethodontid salamanders in Nuclear Central America, a global hotspot of diversity and endemism, are especially vulnerable to warming and altered precipitation regimes. We applied species distribution models (SDM) to project the current and future distributions of 35 salamander species under moderate (SSP2-4.5) and high-emissions (SSP5-8.5) climate scenarios. We quantified changes in range size, species–area relationships, phylogenetic diversity, endemism, and evolutionary distinctiveness, and evaluated biodiversity patterns across protected areas under different governance

☐ <https://link.springer.com/article/10.1007/s10531-026-03346-4>

【生态学核心】

10. Warming Reduces the Positive Effect of Nitrogen Addition on Soil Organic Carbon in Grasslands

☐ Ecology Letters | Mon, 06 Apr 2026 18:

☐ <https://onlinelibrary.wiley.com/doi/10.1111/ele.70379>

【环境与气候】

11. Three decades of continuous warming in temperate forests destabilizes persistent forms of soil organic matter

☐ Science of the Total Environment

□ Abstract:

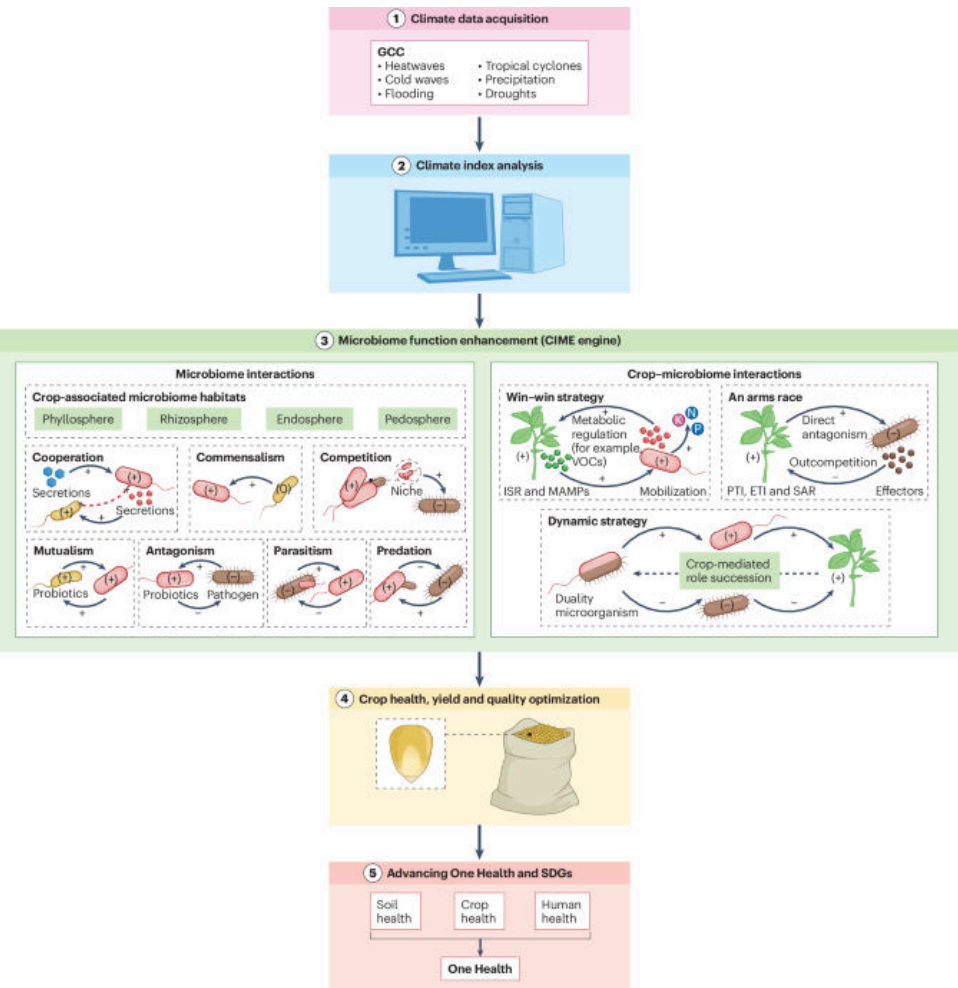
Publication date: 10 May 2026Source: Science of The Total Environment, Volume 1029Author(s): Atzín X. San Román, Serita D. Frey, Melissa A. Knorr, Huan Tong, Jerry M. Melillo, Myrna J. Simpson

□ <https://www.sciencedirect.com/science/article/pii/S0048969726004419>

【植物与农业】

12. Microbiome-driven innovations for climate-resilient crop production

□ Nature Food | 2026-04-03



□ Abstract:

Nature Food, Published online: 03 April 2026; doi:10.1038/s43016-026-01337-wCrop productivity under climate stress remains constrained by conventional agricultural approaches that underuse plant – microbiome interactions. A microbiome-centred, climate-responsive framework was proposed to enhance crop resilience and agro-sustainability by prioritizing targeted manipulation of crop-associated microbiomes, offering a scalable and adaptive pathway to buffer climate stresses and stabilize crop performance.

□ <https://doi.org/10.1038/s43016-026-01337-w>

13. Simultaneous Nitrogen Addition and Warming Cause More Plant Species Loss Than Single Factors Alone

☐ Journal of Vegetation Science | Fri, 03 Apr 2026 22:

☐ Abstract:

Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026.

☐ <https://onlinelibrary.wiley.com/doi/10.1111/jvs.70135>

【土壤与生物地球化学】

14. Estimating particulate organic matter flux from in-situ optics: A framework for correcting for suspended particles and incorporating depth-dependent degradation

☐ Biogeosciences | Tue, 31 Mar 2026 08:

☐ Abstract:

Estimating particulate organic matter flux from in-situ optics: A framework for correcting for suspended particles and incorporating depth-dependent degradation Nasrollah Moradi, Lili Hufnagel, Simon Ramondenc, Clara M. Flintrop, Rainer Kiko, Tim Fischer, Helena Hauss, Arne Körtzinger, Gerhard Fischer, and Morten H. Iversen Biogeosciences, 23, 2179–2203, <https://doi.org/10.5194/bg-23-2179-2026>, 2026 The presented framework advances estimates of particulate organic carbon and nitrogen fluxes in the water column derived from in situ imaging-based particle size distribution data. By accounting for non-sinking particles, depth-dependent degradation, and a dynamic carbon-to-nitrogen ratio that evolves with particle age during settling, it provides a more accurate representation of organic matter

☐ <https://doi.org/10.5194/bg-23-2179-2026>

15. Soil Preservation in Warming Climate: Organic Amendments Enhance Microbial Carbon Use Efficiency in Mediterranean Soils

☐ European Journal of Soil Science | Mon, 06 Apr 2026 22:

☐ Abstract:

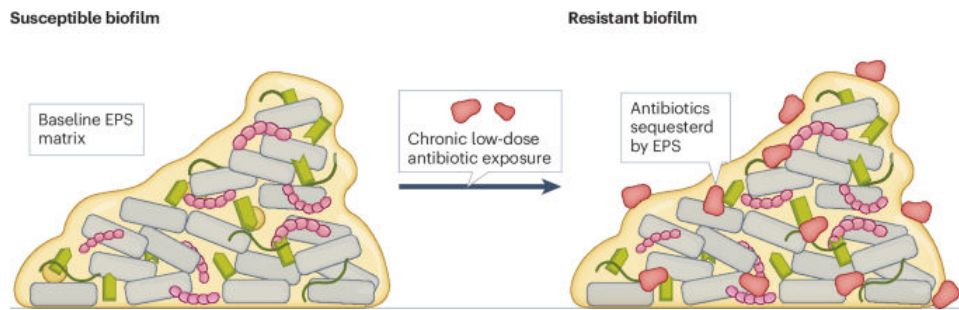
European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

☐ <https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70323>

【海洋与水生】

16. Catching antibiotics outside the cells

☐ Nature Water | 2026-03-30



□ Abstract:

Nature Water, Published online: 30 March 2026; doi:10.1038/s44221-026-00609-wAntibiotics are increasingly being detected in water systems worldwide. Long-term exposure to low-dose antibiotics causes *Escherichia coli* to remodel its extracellular matrix, limiting antibiotic access to the bacterial cells.

□ <https://doi.org/10.1038/s44221-026-00609-w>

【生态学核心】

17. Late Neolithic ‘decline’ is a grave question

□ Nature Ecology & Evolution | 2026-04-03



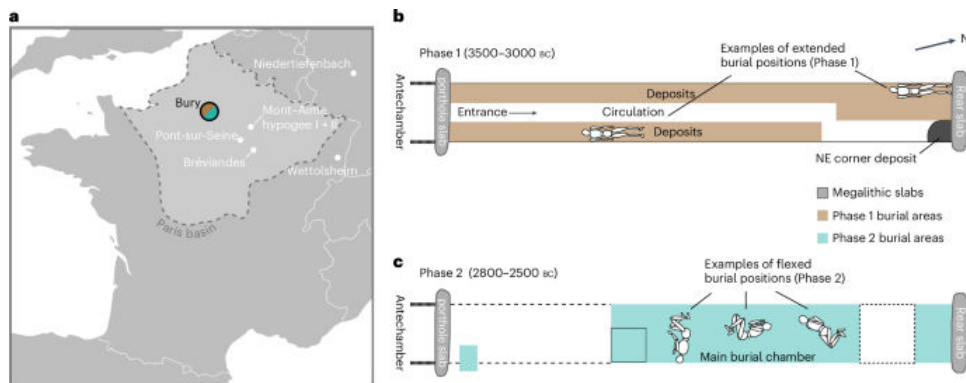
□ Abstract:

Nature Ecology & Evolution, Published online: 03 April 2026; doi:10.1038/s41559-026-03007-3Were there catastrophic population collapses across Europe in the late Neolithic and, if so, what factors were responsible? Ancient DNA from 133 people whose remains were interred in a megalithic gallery grave may shed light on this question.

☐ <https://doi.org/10.1038/s41559-026-03007-3>

18. Population discontinuity in the Paris Basin linked to evidence of the Neolithic decline

☐ Nature Ecology & Evolution | 2026-04-03



☐ Abstract:

Nature Ecology & Evolution, Published online: 03 April 2026; doi:10.1038/s41559-026-03027-z Analysis of ancient human genomes from a Neolithic collective gallery grave in northern France points to population discontinuity and turnover between the third and fourth millennium BC.

☐ <https://doi.org/10.1038/s41559-026-03027-z>

【环境与气候】

19. Invasive Burmese pythons (*Python bivittatus*) as sentinels for PFAS biomonitoring in the Greater Everglades Ecosystem

☐ Science of the Total Environment

☐ Abstract:

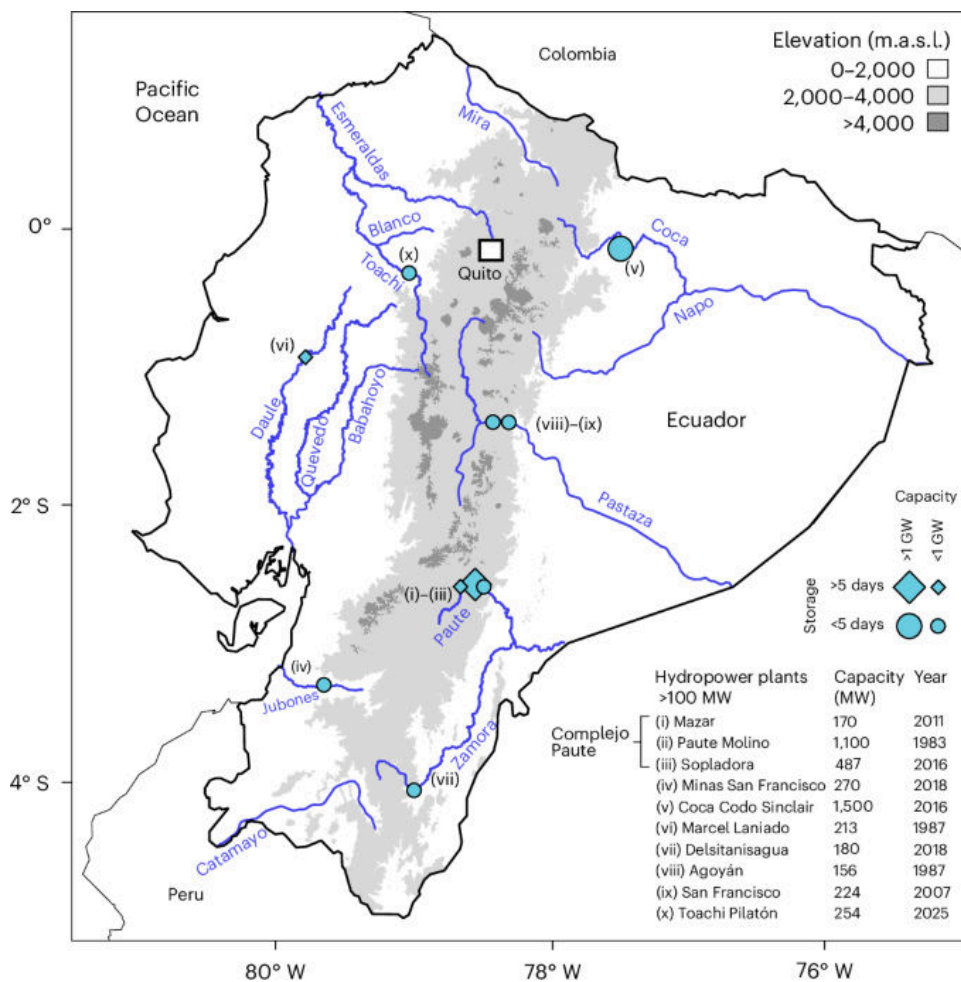
Publication date: 10 May 2026 Source: Science of The Total Environment, Volume 1029 Author(s): Lauren E. Blackman, Qaim Mehdi, Alina Timshina, Ian A. Bartoszek, Melissa A. Miller, Ian C. Easterling, Kevin Olejniczak, Joe Aufmuth, John A. Bowden

☐ <https://www.sciencedirect.com/science/article/pii/S0048969726004262>

【海洋与水生】

20. Variable renewables fortify Ecuador's power system against recurrences of drought-driven energy crises

☐ Nature Water | 2026-04-07



□ Abstract:

Nature Water, Published online: 07 April 2026; doi:10.1038/s44221-026-00617-w Ecuador's reliance on hydropower led to an energy crisis following two failed rainy seasons, highlighting the need for alternative solutions. Here the authors demonstrate that integrating solar and wind power with hydropower, leveraging 'extreme-year synergy', can stabilize energy supply during droughts, offering a model for increased power system resilience in hydro-dependent regions globally.

□ <https://doi.org/10.1038/s44221-026-00617-w>

【综合顶刊】

21. DefensePredictor: A machine learning model to discover prokaryotic immune systems

□ Science | 2026-04-02T07:00:00Z

□ <https://doi.org/10.1126/science.adv7924>

【遥感与模型】

22. Improved Joint Estimation of Daily Actual Evapotranspiration and Gross Primary Production by Integrating Landsat, Sentinel-2, Sentinel-1, and Climate Data With Machine Learning

□ Journal of Geophysical Research - Biogeosciences | Sun, 05 Apr 2026 22:

□ Abstract:

Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 4, April 2026.

□ <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JG009246>

【生态学核心】

23. Beyond Biomass: How Interactions Shape Species' Contribution to Ecosystem Functioning

□ Ecology Letters | Mon, 06 Apr 2026 17:

□ <https://onlinelibrary.wiley.com/doi/10.1111/ele.70370>

24. Protist-Bacteria Trophic Interactions Contribute to Higher Community Stability in Subsoils Than Topsoils

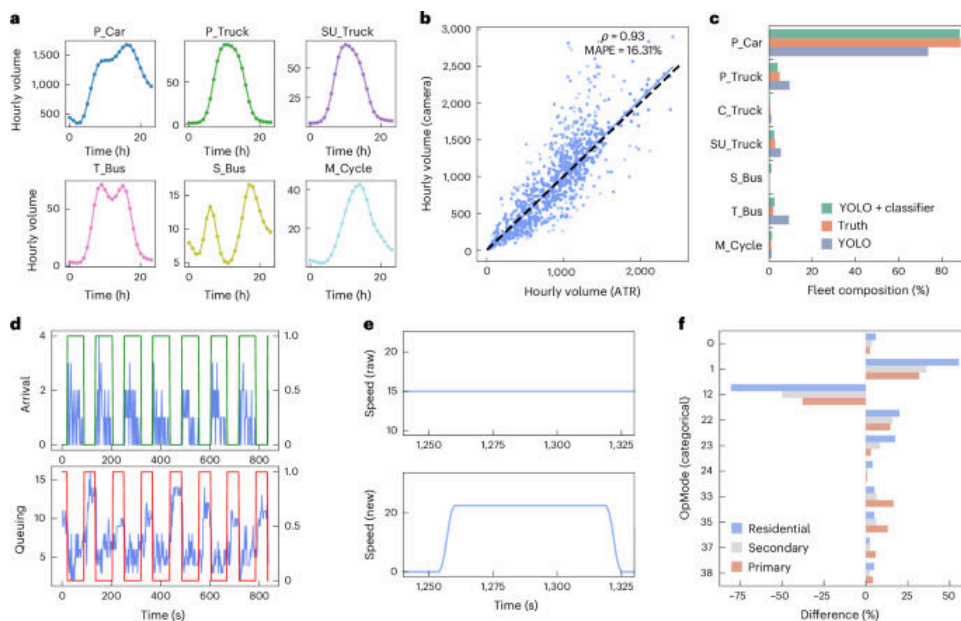
□ Global Change Biology | Mon, 06 Apr 2026 05:

□ <https://onlinelibrary.wiley.com/doi/10.1111/gcb.70839>

【环境与气候】

25. Ubiquitous data-driven framework for traffic emission estimation and policy evaluation

□ Nature Sustainability | 2026-04-02



□ Abstract:

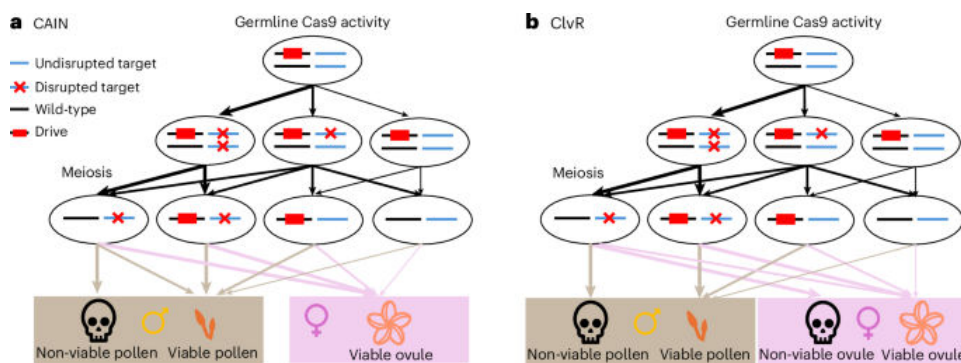
Nature Sustainability, Published online: 02 April 2026; doi:10.1038/s41893-026-01797-9 Urban traffic emission estimates are often too coarse or slow to evaluate decarbonization policies. A study now fuses data from traffic cameras and mobile phones to estimate citywide emissions in near-real time, finding strong spatiotemporal variation and a 16–22% emission drop after congestion pricing.

□ <https://doi.org/10.1038/s41893-026-01797-9>

【植物与农业】

26. Seed dormancy shapes gene drive dynamics in plants

□ Nature Plants | 2026-04-03



□ Abstract:

Nature Plants, Published online: 03 April 2026; doi:10.1038/s41477-026-02256-1 The authors present a comprehensive plant-specific modelling framework for CRISPR gene drives: dormant seed banks can slow spread and require larger releases but can also ease weed elimination and limit unintended spread to nearby populations.

□ <https://doi.org/10.1038/s41477-026-02256-1>

【土壤与生物地球化学】

27. Stochastic Processes Dominate Prokaryotic Community Assembly Yet Decouple From Nitrogen Cycling in Mediterranean Soils

☐ European Journal of Soil Science | Mon, 06 Apr 2026 22:

☐ Abstract:

European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

☐ <https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70311>

【遥感与模型】

28. Atmospheric Ammonia Deposition: A Significant Source of Nitrogen to an Oligotrophic Lake and Its Watershed

☐ Journal of Geophysical Research - Biogeosciences | Sat, 04 Apr 2026 05:

☐ Abstract:

Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 4, April 2026.

☐ <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JG009389>

【海洋与水生】

29. Probabilistic storm surge forecasting in the Bohai Sea: A deep learning framework with adaptive uncertainty quantification

☐ Estuarine Coastal and Shelf Science

☐ Abstract:

Publication date: 1 August 2026Source: Estuarine, Coastal and Shelf Science, Volume 336Author(s): Changyu Su, Miaohua Mao

☐ <https://www.sciencedirect.com/science/article/pii/S0272771426001721>

【综合顶刊】

30. Atomic-resolution imaging of gold species at organic liquid-solid interfaces

☐ Science | 2026-04-02T06:00:11Z

☐ Abstract:

Science, Volume 392, Issue 6793, Page 77-82, April 2026.

☐ <https://doi.org/10.1126/science.adw2469>

31. Rising flood risks in semiarid South Asia driven by changing intraseasonal oscillations under global warming

☐ Science Advances | 2026-04-03T07:00:00Z

☐ <https://doi.org/10.1126/sciadv.aea0082>

32. Regulating interface electric field to stabilize high-voltage KVPO4F positive electrode for sustainable potassium-ion batteries

☐ Nature Communications | 2026-04-07

☐ Abstract:

Nature Communications, Published online: 07 April 2026; doi:10.1038/s41467-026-71647-x Potassium-ion batteries face stability issues at high voltages. Here, authors use nitrogen-doped nanotubes to regulate the interfacial electric field of a polyanionic KVPO4F positive electrode. This stabilizes the structure and interface, enabling a specific energy of 454.8 Wh kg⁻¹ and a cycling lifespan over 2000 cycles.

☐ <https://doi.org/10.1038/s41467-026-71647-x>

【生态学核心】

33. Innovative Behaviour Is Associated With Increased Collision Incidence in Birds

☐ Ecology Letters | Fri, 03 Apr 2026 20:

☐ <https://onlinelibrary.wiley.com/doi/10.1111/ele.70383>

34. Sunflower Sea Stars Can Counteract Warming-Driven Increases in Sea Urchin Populations

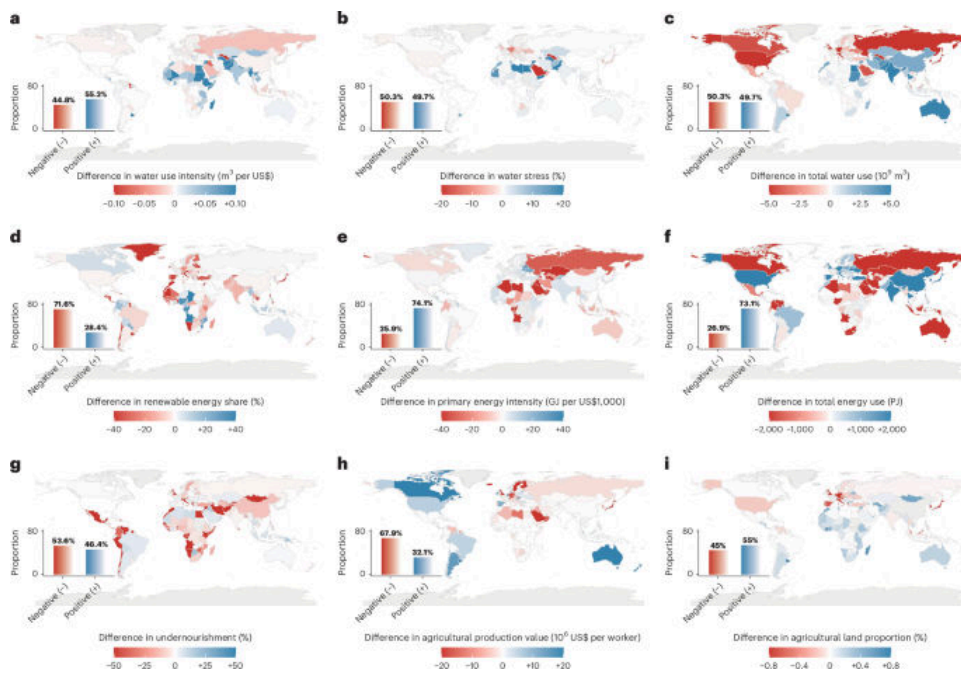
☐ Global Change Biology | Thu, 02 Apr 2026 23:

☐ <https://onlinelibrary.wiley.com/doi/10.1111/gcb.70848>

【环境与气候】

35. The dual impact of trade on the water–energy–food nexus globally

☐ Nature Sustainability | 2026-04-02



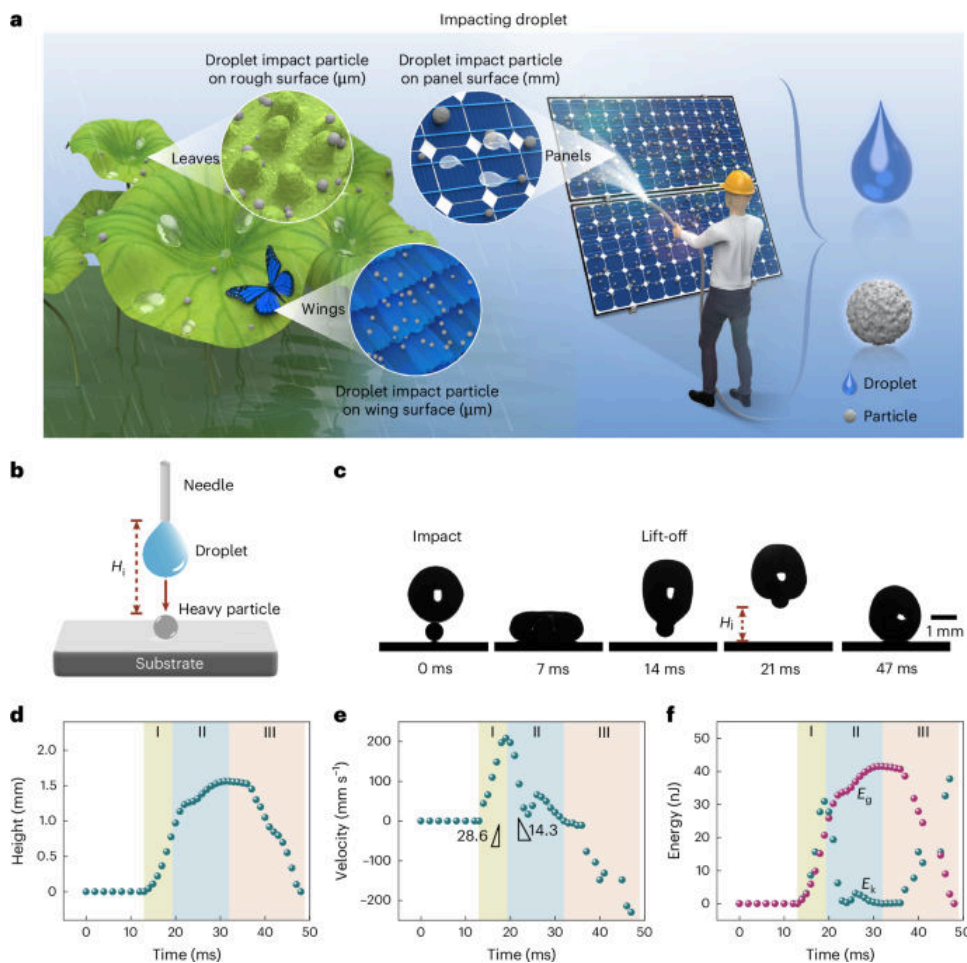
☐ Abstract:

Nature Sustainability, Published online: 02 April 2026; doi:10.1038/s41893-026-01796-w Global trade tightly links water, energy and food systems, but its effects on cross-border sustainability and equity remain unclear. A study now maps global water – energy – food trade networks, finding that trade balances overall resource gains while deepening cross-country income inequalities.

☐ <https://doi.org/10.1038/s41893-026-01796-w>

36. Liquid droplet mops

☐ Nature Sustainability | 2026-04-02



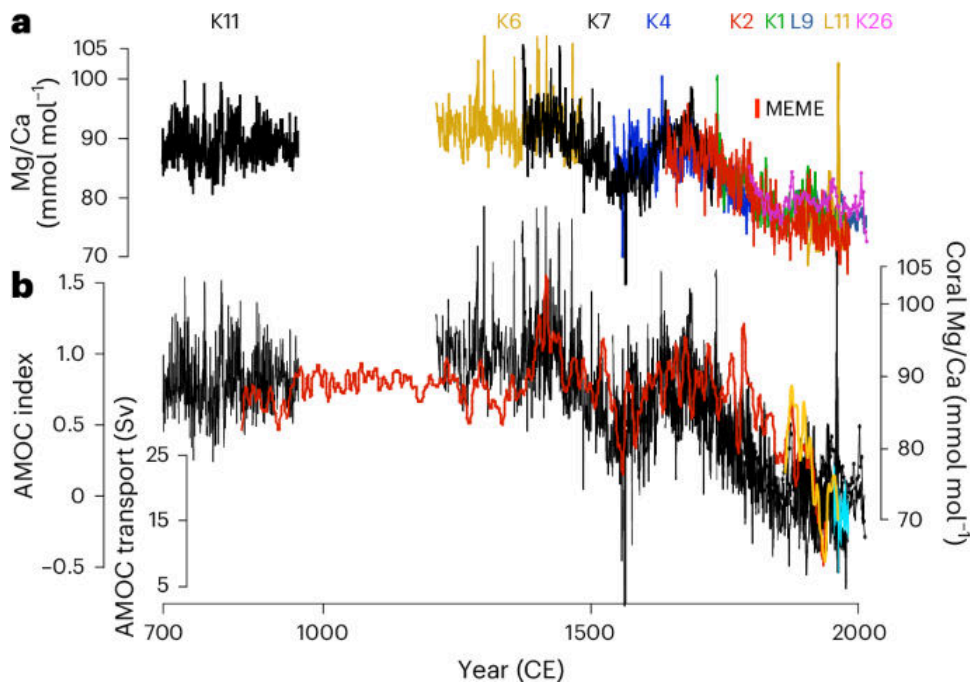
□ Abstract:

Nature Sustainability, Published online: 02 April 2026; doi:10.1038/s41893-026-01804-z Surface cleaning is needed in various industrial systems but remains water intensive due to low water utilization efficiency. This study proposes a method—liquid droplet mops—that can efficiently clean superhydrophobic-coated solar panels with substantially reduced water consumption.

□ <https://doi.org/10.1038/s41893-026-01804-z>

37. Millennial-scale Atlantic overturning circulation led by the Southern Ocean

□ Nature Geoscience | 2026-04-02



☐ Abstract:

Nature Geoscience, Published online: 02 April 2026; doi:10.1038/s41561-026-01959-6 Changes in Southern Ocean circulation have preceded shifts in the Atlantic Meridional Overturning Circulation in the North Atlantic over the past 1,300 years, according to an analysis of bamboo coral records from the southwest Pacific.

☐ <https://doi.org/10.1038/s41561-026-01959-6>

38. Climate impacts apple pollination, yield and economic outcomes of farmers

☐ Science of the Total Environment

☐ Abstract:

Publication date: 10 May 2026 Source: Science of The Total Environment, Volume 1029 Author(s): Clémence Riva, Stella Gachoki, Noureddine Adjlane, Danilo Bevk, Yamina Haider, Fani Hatjina, Šimun Kolega, Marin Kovačić, Ioannis Manthos, Zlatko Puškadija, Thomas Sotiropoulos, Bojan Stipešević, Fabrice Requier

☐ <https://www.sciencedirect.com/science/article/pii/S0048969726004316>

【海洋与水生】

39. Invertebrate community structure and spatial variation associated with *Thalassodendron ciliatum* beds in Tomia Island, Wakatobi National Park

☐ Estuarine Coastal and Shelf Science

☐ Abstract:

Publication date: 1 August 2026 Source: Estuarine, Coastal and Shelf Science, Volume 336 Author(s): Andrea Sofia Aleman-Garcia, Rohani Ambo-Rappe, Inayah Yasir, Yuyu La Nafie

☐ <https://www.sciencedirect.com/science/article/pii/S0272771426001794>

【保护与生物多样性】

40. Scaling biodiversity conservation through institutional reform

☐ Conservation Biology | Tue, 31 Mar 2026 22:

☐ Abstract:

Conservation Biology, EarlyView.

☐ <https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70282>

【植物与农业】

41. Occupancy-based mechanism is the chief mode of ROS1 function in preventing DNA hypermethylation

☐ Nature Plants | 2026-04-02

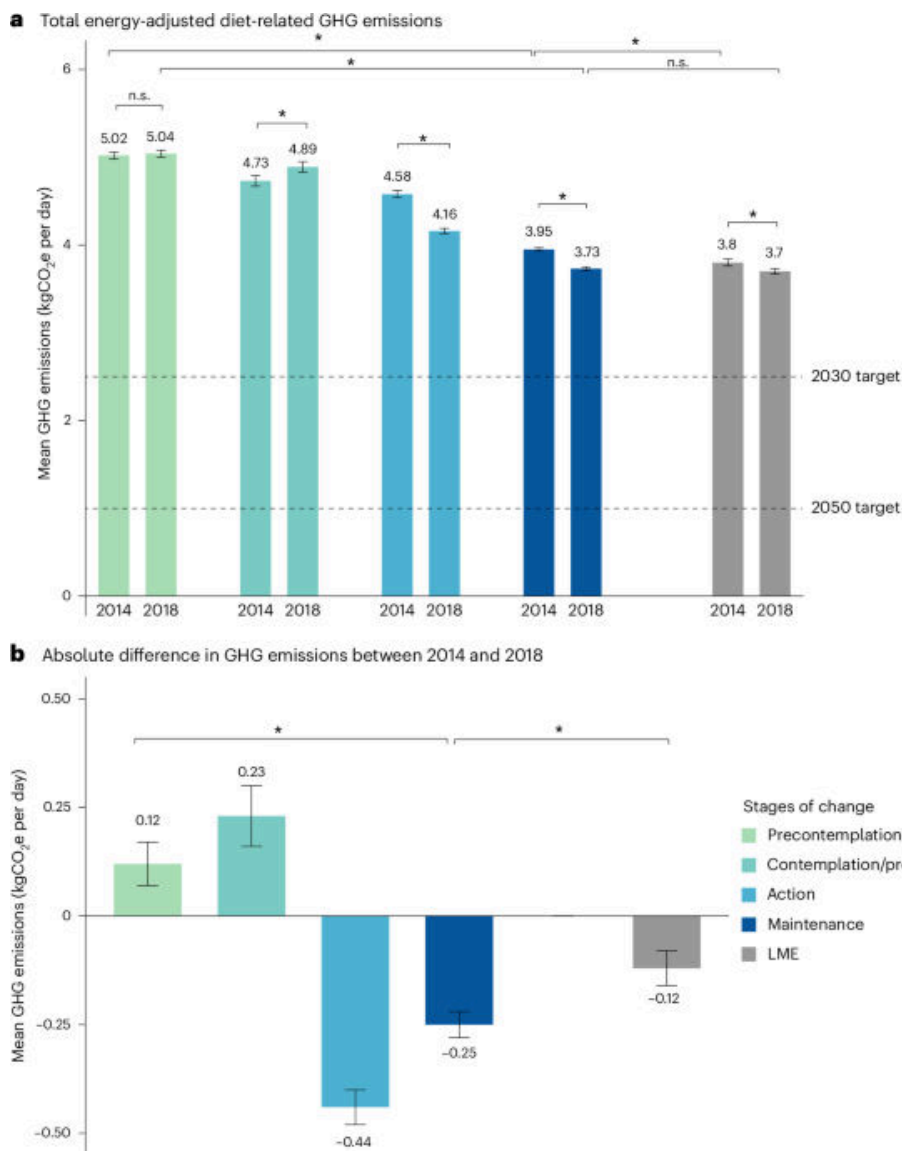
☐ Abstract:

Nature Plants, Published online: 02 April 2026; doi:10.1038/s41477-026-02258-z This study shows that ROS1 mediates DNA demethylation mainly via an occupancy-based passive mechanism, challenging the traditional active model, and reveals that ROS1 occupancy marks and regulates accessible chromatin, highlighting its diverse roles.

☐ <https://doi.org/10.1038/s41477-026-02258-z>

42. Greater readiness to reduce meat consumption is associated with lower greenhouse gas emissions

☐ Nature Food | 2026-04-06



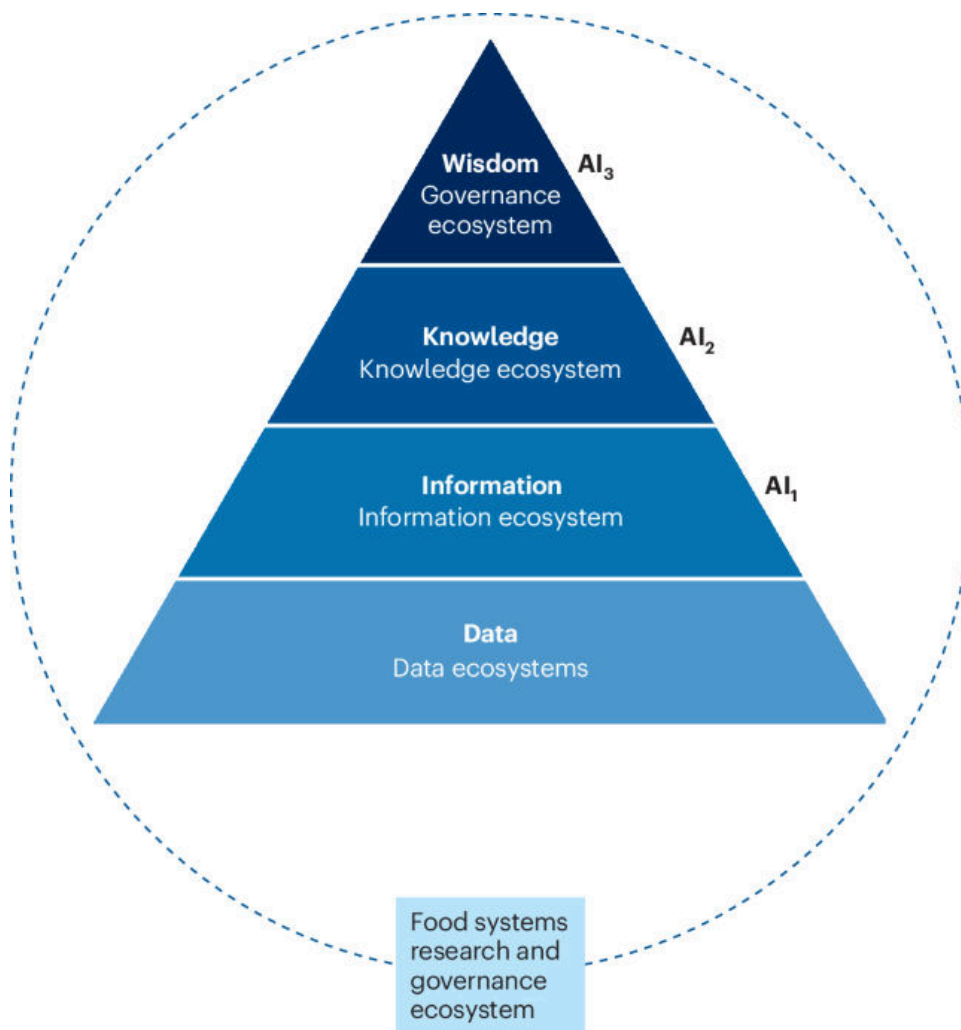
☐ Abstract:

Nature Food, Published online: 06 April 2026; doi:10.1038/s43016-026-01332-1 Greater readiness to reduce meat consumption was associated with actual decreases in meat intake and associated greenhouse gas emissions among a cohort of French adults, primarily through reduced ruminant meat consumption, with smaller declines in pork and poultry. However, these reductions remain insufficient to meet national climate targets.

☐ <https://doi.org/10.1038/s43016-026-01332-1>

43. Artificial intelligence and the foundations of food systems research

☐ Nature Food | 2026-04-02



☐ Abstract:

Nature Food, Published online: 02 April 2026; doi:10.1038/s43016-026-01339-8 Artificial intelligence has recently become an invisible infrastructure shaping how knowledge is produced and applied in food systems. From data pipelines to policy modelling, artificial intelligence-mediated workflows are becoming integral to research practice. An analysis of the intersection of agricultural economics, food systems and digitalization suggests that this transformation is reshaping scientific authority, professional competence and governance in ways that demand deliberate institutional design.

☐ <https://doi.org/10.1038/s43016-026-01339-8>

44. The Alpine Germination Syndrome Differs Between Temperate and Mediterranean Habitats

☐ Journal of Vegetation Science | Fri, 03 Apr 2026 01:

☐ Abstract:

Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026.

☐ <https://onlinelibrary.wiley.com/doi/10.1111/jvs.70134>

45. Effects of Grain Size on Plant Diversity Patterns in a Mediterranean Mountain

☐ Journal of Vegetation Science | Fri, 03 Apr 2026 00:

☐ Abstract:

Journal of Vegetation Science, Volume 37, Issue 2, March/April 2026.

☐ <https://onlinelibrary.wiley.com/doi/10.1111/jvs.70133>

【土壤与生物地球化学】

46. Who Do You Strive To Be? Exploring Norms of Good Farming for Sustainable Soil Management in Four European Regions

☐ European Journal of Soil Science | Mon, 06 Apr 2026 22:

☐ Abstract:

European Journal of Soil Science, Volume 77, Issue 2, March–April 2026.

☐ <https://bsssjournals.onlinelibrary.wiley.com/doi/10.1111/ejss.70318>

【遥感与模型】

47. Drivers of Dissolved Organic Matter Quality and Concentration in a Mountainous Subarctic Watershed, Yukon, Canada

☐ Journal of Geophysical Research - Biogeosciences | Fri, 03 Apr 2026 02:

☐ Abstract:

Journal of Geophysical Research: Biogeosciences, Volume 131, Issue 4, April 2026.

☐ <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025JG009320>

【综合顶刊】

48. IL-22 from enteroendocrine cells promotes early-life gut motility in zebrafish through the microbiota

☐ Science | 2026-04-02T06:00:11Z

☐ Abstract:

Science, Volume 392, Issue 6793, Page 69-76, April 2026.

☐ <https://doi.org/10.1126/science.adr1707>

49. Global mapping of disaggregated international trade-linked transportation CO2 emissions

☐ Science Advances | 2026-04-03T07:00:00Z

☐ <https://doi.org/10.1126/sciadv.adz1670>

50. Intrinsic attraction driving the high temperature performance of additively manufactured aluminum alloys

☐ Nature Communications | 2026-04-07

☐ Abstract:

Nature Communications, Published online: 07 April 2026; doi:10.1038/s41467-026-71390-3 This study introduces an intrinsic attraction strategy to enhance the high-temperature performance of additively manufactured aluminum alloys. By confining atomic defects and stabilizing precipitates, the alloy retains 305 MPa at 300 °C with creep resistance up to 400 °C.

☐ <https://doi.org/10.1038/s41467-026-71390-3>

【生态学核心】

51. The Separate Roles of Vascular Plants and Sphagnum Mosses in Regulating the Net CO2 Exchange in a Boreal Peatland During Key Phenological Phases

☐ Global Change Biology | Fri, 03 Apr 2026 03:

☐ <https://onlinelibrary.wiley.com/doi/10.1111/gcb.70834>

【综合顶刊】

52. Toward AI foundation models for epidemics: Promise, challenges, and paths forward

☐ PNAS | 2026-03-13T07:00:00Z

☐ Abstract:

Foundation models—large AI systems pretrained on broad, heterogeneous data—are transforming scientific discovery. These models (e.g., GPT, GenCast, AlphaFold) excel at learning generalizable representations and adapting to new tasks with limited data. Yet,...

☐ <https://www.pnas.org/doi/abs/10.1073/pnas.2526192123>

53. Pedro A. Sanchez (1940–2026), a social soil scientist

☐ PNAS | 2026-03-20T07:00:00Z

☐ Abstract:

Cuban-born Pedro A. Sanchez (1940-2026) became the face of tropical soil science at Cornell University and the North Carolina State University. His combination of process-based understanding of soils, emphasis on farmer choices and conducive policy ...

☐ <https://www.pnas.org/doi/abs/10.1073/pnas.2604417123>

54. The archaeal roots of eukaryotic life

☐ PNAS | 2026-03-13T07:00:00Z

☐ Abstract:

Resolving the biological and geological events that led to the origin of eukaryotes is an ongoing challenge in biology. A major step in the evolution of complex cellular life was the merger between an ancestral host cell and a bacterium (that became the ...

☐ <https://www.pnas.org/doi/abs/10.1073/pnas.2516062123>

【环境与气候】

55. [ASAP] Amine-Containing Micropollutants Exposure Reshapes Sludge Anaerobic Digestion via Enzymatic Inhibition and Stress-Mediated Alteration of Methanogenic Pathways

☐ Environmental Science & Technology | Tue, 07 Apr 2026 02:

☐ Abstract:

Environmental Science & TechnologyDOI: 10.1021/acs.est.5c10074

☐ <https://doi.org/10.1021/acs.est.5c10074>

56. [ASAP] Beyond Metal(loid) Immobilization: Redox-Stratified Biocrusts Shield Humid Mining Regions

☐ Environmental Science & Technology | Tue, 07 Apr 2026 03:

☐ Abstract:

Environmental Science & TechnologyDOI: 10.1021/acs.est.5c13821

☐ <https://doi.org/10.1021/acs.est.5c13821>

57. [ASAP] Presence of Legacy and Emerging PFAS in Human Liver Specimens Banked in the United States from 2000 to 2024

☐ Environmental Science & Technology | Tue, 07 Apr 2026 06:

☐ Abstract:

Environmental Science & Technology DOI: 10.1021/acs.est.5c14780

☐ <https://doi.org/10.1021/acs.est.5c14780>

【海洋与水生】

58. Poor water quality may reverse protective effects of blue space on metabolic health

☐ Nature Water | 2026-03-31

☐ Abstract:

Nature Water, Published online: 31 March 2026; doi:10.1038/s44221-026-00618-9 While blue space has been linked to metabolic health, research has primarily focused on water quantity rather than quality. This study finds that lower water coverage and higher exposure to black-odorous water increase risk of metabolic syndrome.

☐ <https://doi.org/10.1038/s44221-026-00618-9>

【保护与生物多样性】

59. A framework for maximizing the benefit from retaining regrowth on private land

☐ Conservation Biology | Sun, 05 Apr 2026 23:

☐ Abstract:

Conservation Biology, EarlyView.

☐ <https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70278>

60. Valuing the contributions of conservation volunteers to coastal society

☐ Conservation Biology | Mon, 30 Mar 2026 23:

☐ Abstract:

Conservation Biology, EarlyView.

☐ <https://conbio.onlinelibrary.wiley.com/doi/10.1111/cobi.70276>

61. Myosin II-driven contractions of supporting cap cells promote sensory adaptation of *Drosophila* proprioceptors

☐ Current Biology

☐ Abstract:

Publication date: 6 April 2026 Source: Current Biology, Volume 36, Issue 7 Author(s): Chonglin Guan, Kengo Nishi, Christian T. Kreis, Oliver Bäumchen, Martin C. Göpfert, Christoph F. Schmidt

☐ <https://www.sciencedirect.com/science/article/pii/S096098222600223X>

【植物与农业】

62. Breeding mitochondria in crops

☐ Nature Plants | 2026-04-02

☐ Abstract:

Nature Plants, Published online: 02 April 2026; doi:10.1038/s41477-026-02275-y Mitochondrial genes in most crops are inherited maternally. In this issue of Nature Plants, Gonzalez-Duran et al. report that mitochondrial genes can be inherited from both parents.

☐ <https://doi.org/10.1038/s41477-026-02275-y>

【环境与气候】

63. Addendum: Sea-level rise at the end of the last deglaciation dominated by North American ice sheets

☐ Nature Geoscience | 2026-04-07

☐ Abstract:

Nature Geoscience, Published online: 07 April 2026; doi:10.1038/s41561-026-01970-x Addendum: Sea-level rise at the end of the last deglaciation dominated by North American ice sheets

☐ <https://doi.org/10.1038/s41561-026-01970-x>

【海洋与水生】

64. Climatic regime and its role in shaping patterns of growth and reproduction in the genus *Callichirus* (Decapoda: Axiidea: Callichiridae)

☐ Estuarine Coastal and Shelf Science

☐ Abstract:

Publication date: 1 August 2026 Source: Estuarine, Coastal and Shelf Science, Volume 336 Author(s): Patricio Hernández, Marcelo M. Rivadeneira

☐ <https://www.sciencedirect.com/science/article/pii/S0272771426001617>

【保护与生物多样性】

65. Big1 is a cell-cycle regulator linking cell size to basal body number in *Tetrahymena thermophila*

☐ Current Biology

☐ Abstract:

Publication date: 6 April 2026
Source: Current Biology, Volume 36, Issue 7
Author(s): Alexander J. Stemm-Wolf, Adam W.J. Soh, Lisa E. Mitchell, Huangqingbo Sun, Erik Collet, Gayatri L. Dholakia, Vikram Raju, Jay R. Hesselberth, J. Matthew Taliaferro, Robert F. Murphy, Lydia R. Heasley, Chad G. Pearson

☐ <https://www.sciencedirect.com/science/article/pii/S0960982226002526>

66. A developmental switch in corticoclaustral signaling

☐ Current Biology

☐ Abstract:

Publication date: 6 April 2026
Source: Current Biology, Volume 36, Issue 7
Author(s): Tarek Shaker, Mohammed Almokdad, Lotte J.A.M. Razenberg, Mahesh M. Karnani, Jesse Jackson

☐ <https://www.sciencedirect.com/science/article/pii/S0960982226002368>

【环境与气候】

67. Determining the influence of water availability for photosynthetic life in Antarctica: a systematic review

☐ Environmental Research Letters | 2026-04-06T23:00:00Z

☐ Abstract:

Antarctica is paradoxically the driest continent on the planet, despite holding most of the Earth's freshwater. In these conditions, water availability governs distribution, health, and survival of vegetation on the continent. This relationship remains poorly understood due to extreme weather, remoteness and logistical challenges. This limits research access and long-term observations which have resulted in a fragmented understanding of how water availability shapes Antarctica's photoautotrophs. This systematic review analysed 113 peer-reviewed studies published between 1990 and 2025 using a PRISMA 2020 approach. The inclusion criteria comprised of studies only in Antarctica where both water availability and photosynthetic life were discussed. Studies went through two screening stages, first

☐ <https://iopscience.iop.org/article/10.1088/1748-9326/ae51ae>

【海洋与水生】

68. Environmental constraints shape growth plasticity, early maturation, and longevity in an endangered freshwater fish

☐ Hydrobiologia | 2026-03-26

☐ Abstract:

The interaction between organisms and their environment strongly influence growth, longevity, and fitness, yet the roles of resource availability and predation remain poorly resolved for threatened freshwater fishes. This study investigates how environmental conditions shape growth, survival, and life-history traits in two spatially isolated populations of the endangered Macquarie perch (*Macquaria australasica*) inhabiting contrasting lotic and lentic habitats in south-eastern Australia. Age, growth, fecundity, and mortality were estimated from otolith and gonadal analyses; predation pressure was assessed through stomach content analysis of *Maccullochella* spp.; and compensatory growth was tested using a 12-month pond grow-out experiment. Fish from Cataract Dam exhibited markedly reduced gro

☐ <https://link.springer.com/article/10.1007/s10750-026-06197-w>

【保护与生物多样性】

69. Assessing direct effects of insect change on insectivore populations in the United Kingdom

☐ Biodiversity and Conservation | 2026-04-06

☐ Abstract:

Declines in insect abundance are a cause for concern, with potential downstream impacts on ecosystem function. Insects are key food resources for insectivorous vertebrates, with evidence that reductions in prey availability may influence population dynamics. Yet quantifying direct effects at large spatial scales is difficult, both due to data limitations and because correlations may reflect shared responses to environmental change rather than trophic links. We assess the role of insect abundance in the dynamics of 10 insectivores (five birds, five bats) in the United Kingdom by pairing insect and vertebrate abundance data at three spatial resolutions (100, 50, 10 km) from multiple citizen science programmes. To address the challenges of inference and combining heterogeneous data, we adopt

☐ <https://link.springer.com/article/10.1007/s10531-026-03329-5>

【海洋与水生】

70. Climate change linked to infertility in well-known shark species

☐ Journal of Fish Biology | Mon, 06 Apr 2026 08:

☐ Abstract:

Journal of Fish Biology, Volume 108, Issue 2, Page 417-418, February 2026.

☐ <https://onlinelibrary.wiley.com/doi/10.1111/jfb.70404>

【遥感与模型】

71. Forecasting bryozoan assemblage dynamics under simulated climate change

☐ Ecological Modelling

☐ Abstract:

Publication date: July 2026 Source: Ecological Modelling, Volume 517 Author(s): Mira Baer, Korinna T. Allhoff, David K.A. Barnes, Franziska Koch

☐ <https://www.sciencedirect.com/science/article/pii/S0304380026001055>

72. Climate change and soybean production: A regional evaluation in the Northwestern part of Türkiye

☐ Ecological Modelling

☐ Abstract:

Publication date: July 2026 Source: Ecological Modelling, Volume 517 Author(s): Bilge Arslan, Şerife Tülin Akkaya Aslan, Burak Şen

☐ <https://www.sciencedirect.com/science/article/pii/S0304380026001134>

【生态学核心】

73. Over three-quarters of earthworm species lack protection in China, a crisis exacerbated by climate change

☐ Ecography | Fri, 03 Apr 2026 06:

☐ Abstract:

Ecography, EarlyView.

☐ <https://nsojournals.onlinelibrary.wiley.com/doi/10.1002/ecog.08554>

【综合顶刊】

74. Rapid summer methane emission decline in high-latitude plains linked to 2021 drought

☐ Communications Earth & Environment | 2026-04-07

☐ Abstract:

Communications Earth & Environment, Published online: 07 April 2026; doi:10.1038/s43247-026-03433-y The 2021 drought across northern high-latitude Eurasia reduced summer methane emissions by 20%, mainly from wetlands due to lower liquid water content, according to an analysis using a greenhouse gas monitoring inversion system and a merged satellite dataset.

☐ <https://doi.org/10.1038/s43247-026-03433-y>

【海洋与水生】

75. Event-driven monitoring reveals multi-pesticide contamination of small standing waters beyond adjacent de facto farming practice risks

☐ Hydrobiologia | 2026-03-20

☐ Abstract:

Agricultural practices are a major source of pesticide contamination in freshwater ecosystems. Lentic small water bodies (LSWBs), which make up the vast majority of freshwater systems, are particularly vulnerable to various pesticide entry routes, including surface runoff, spray drift and subsurface flow from adjacent agricultural land. This study aimed to assess whether event-driven monitoring could effectively detect pesticide contamination in LSBWs. Using 1 year of event-driven monitoring data from three LSBWs, we compared measured pesticide concentrations with predicted values from the SYNOPS-WEB risk assessment tool. The results revealed notable discrepancies. While SYNOPS-WEB predicted certain contamination events, on-site monitoring frequently showed the presence of pesticides that

☐ <https://link.springer.com/article/10.1007/s10750-026-06130-1>

【土壤与生物地球化学】

76. Can coarse woody material account for the nitrogen imbalance at Hubbard Brook Experimental Forest, New Hampshire, USA?

☐ Biogeochemistry | 2026-03-24

☐ Abstract:

The decay of coarse woody litter serves a potentially important role in forest nitrogen cycling. The carbon-rich, nitrogen-poor chemistry of wood allows it to immobilize, store, and in later stages of decomposition, supply nitrogen to forest ecosystems. The decay of woody litter happens over decadal time scales, making direct observations of its importance to nitrogen cycling challenging. Modeling woody litter decay can provide insights into its role in nitrogen cycling but is complex because it is influenced by microbial stoichiometric demands, wood chemistry, time spent as standing versus downed wood, input rates from mortality and disturbances, decay rates, and whether these processes are dynamic over time. One ecosystem where these uncertainties are particularly relevant is the Hubbard

☐ <https://link.springer.com/article/10.1007/s10533-026-01314-2>

【海洋与水生】

77. Thermal stress reveals contrasting bleaching and trophic patterns in two Southwestern Atlantic octocorals

☐ Coral Reefs | 2026-04-01

☐ Abstract:

Most shallow-water octocorals engage in a symbiotic relationship with photosynthetic dinoflagellates of the family Symbiodiniaceae. Their inherently mixotrophic nature, combining autotrophic and heterotrophic feeding, provides a flexible energy acquisition that supports their metabolic demands. This nutritional plasticity depends on the functional integrity of symbiosis, which can be disrupted by thermal stress, potentially disturbing oxidative balance and causing symbiosis breakdown (i.e., bleaching). Although temperature effects are well known in scleractinian corals, octocoral ecophysiological responses remain poorly understood, especially in South Atlantic reefs. In this study, we used two octocoral species, *Neospongodes atlantica* and *Plexaurella grandiflora*, to test the hypotheses tha

☐ <https://link.springer.com/article/10.1007/s00338-026-02856-5>

【遥感与模型】

78. Downscaling of satellite passive microwave brightness temperature through super-resolution reconstruction

☐ Remote Sensing of Environment

☐ Abstract:

Publication date: June 2026Source: Remote Sensing of Environment, Volume 339Author(s): Jiaxin Chen, Ji Zhou, Tao Zhang, Shaojie Zhao, Ruyin Cao, Jin Ma, Wenbin Tang, Lin Feng

☐ <https://www.sciencedirect.com/science/article/pii/S0034425726001756>

【预印本】

79. Temperature stress resilience in polar *Chlamydomonas* is regulated by acclimation to light and salinity: implications for survival in a changing world

☐ bioRxiv Plant Biology | 2026-04-07

☐ Abstract:

Aquatic algae are key primary producers in the Arctic and Antarctic, yet how cold-water species respond to environmental change is poorly understood. The Polar Regions are increasingly exposed to frequent heat waves, leading to declining ice cover, increased light availability, and decreasing salinity in polar waters. We compared three phylogenetically related but geographically distant polar *Chlamydomonas* species to test how habitat history shapes algal responses to light, salinity, and temperature stress. We assessed the growth, morphology, and

photochemistry of psychrophilic *Chlamydomonas* acclimated to native-like (lower light, higher salinity) and climate-shifted conditions (higher light, lower salinity). Next, we exposed acclimated cultures to a lethal heat shock and observed how accl

□ <https://doi.org/10.1101/2026.04.03.716389>

【植物与农业】

80. Root-soil interactions a key driver of aggregate formation and stability: a trait-based and mechanistic review

□ Plant and Soil | 2026-04-02

□ Abstract:

Soil aggregation controls the interconnections between pores, water retention, and physical protection of soil organic carbon in both micro- and macro-aggregate levels. Although microbial activity and organic matter have been considered, there is mounting evidence that plant roots contribute to aggregate formation and stability by being physically entangled, rhizodeposited, and through an interface between minerals and organic matter termed the rhizosphere. This review synthesized the mechanistic and trait-based pathways by which roots drive soil aggregation across diverse soils and environmental stressors. In addition, we precisely discussed how root-soil interactions perform under various environmental stressors. Furthermore, the impacts of different agricultural management interventions

□ <https://link.springer.com/article/10.1007/s11104-026-08453-3>

81. Bison grazing increases grass foliar nutrients and drought tolerance at fine spatial scales

□ Plant and Soil | 2026-04-02

□ Abstract:

Aims The American bison is often recognized as an ecological engineer because their grazing and wallowing behavior increases plant diversity and productivity. These disturbances increase spatial heterogeneity and can alter plant nutrient and water dynamics. We assessed how bison grazing and wallowing affect grass chemistry and physiology, and whether grass evolutionary legacies mediate species-specific responses. **Methods** We measured grass drought tolerance, photosynthetic activity, and foliar and soil chemistry in four grass species at Konza Prairie Biological Station (Kansas, USA). Species included two dominant C4 panicoids, one C3 panicoid, and one C4 chloridoid species that becomes dominant in grazed areas. We sampled grasses in grazed areas near and away from bison wallows and adjacent

□ <https://link.springer.com/article/10.1007/s11104-026-08422-w>

【保护与生物多样性】

82. On the road from deserts to forests: spatiotemporal modelling of owl road mortality in Israel

□ Biodiversity and Conservation | 2026-04-07

☐ Abstract:

Road networks are an increasing source of wildlife mortality and owls (Strigiformes) are especially vulnerable, because their low-altitude foraging and frequent use of road edges bring them into close contact with vehicles. We analysed 2,631 owl mortality reports collected across Israel between 2008 and 2024, derived from public submissions and routine monitoring, which were validated by Israel Nature and Parks Authority (INPA) rangers. Among 1,014 cases with identified anthropogenic causes, 65.8% were attributed to vehicle collisions, making roadkill the most frequent identified anthropogenic cause in eight of the ten species. Species-specific temporal models provided little support for consistent long-term trends in road mortality; most species showed pronounced interannual variability,

☐ <https://link.springer.com/article/10.1007/s10531-026-03342-8>

【植物与农业】

83. FarmSTEPS, a model for spatial-temporal exploration of diversified and sustainable crop sequence plans for crop farms

☐ Agronomy for Sustainable Development | 2026-03-17

☐ Abstract:

Crop diversification could enhance economic and environmental sustainability of crop farming, but achieving this requires careful planning that accounts for crop interrelationships in a rotation or spatial layout. Several models have been devised to design diversified crop rotations or crop spatial layouts, but they often fail to capture farm settings, production constraints, and sustainability objectives specific to a farm at both temporal and spatial scales. To fill this gap, we developed a novel rule-based whole-farm planning model, FarmSTEPS, for systematically exploring diversified and sustainable farm crop sequence plans in a spatial-temporally explicit manner. FarmSTEPS comprises multiple farm characterization modules, an algorithm for generating farm crop sequence plans, and a mult

☐ <https://link.springer.com/article/10.1007/s13593-026-01097-8>

84. Rhizosphere effects on soil fungal community structure and network stability vary along altitudinal gradients in alpine forests of eastern Tibet

☐ Plant and Soil | 2026-04-02

☐ Abstract:

Aims Soil fungi are critical for ecosystem function maintenance, but how fungal communities respond to environmental factors across dual altitude-rhizosphere scales remains unclear. **This study explores** rhizosphere effects on the variations and driving factors of soil fungal communities along altitude gradients. **Methods** We studied the soil fungal diversity, composition, and co-occurrence network, as well as soil properties in rhizosphere and bulk soil of *Abies georgei* var. *smithii* along three altitudes (3800, 4100, 4400 m) on Shergyla Mountain, eastern Tibet. **Results** Soil fungal richness first increased and then decreased with increasing altitude. The fungal community composition also showed a marked altitudinal divergence, with *Cortinarius* and *Russula* dominating at high and low altitudes,

☐ <https://link.springer.com/article/10.1007/s11104-026-08522-7>

【土壤与生物地球化学】

85. Mobilized dissolved organic matter provides niche space for prokaryotes in the deep subterranean estuary of a sandy beach

☐ Biogeochemistry | 2026-03-17

☐ Abstract:

A fraction of organic carbon in beach sands is mobilizable as dissolved organic matter (DOM). The molecular composition of this mobilizable DOM and its relation to microbial metabolism is central for biogeochemical processes in beach ecosystems, yet, still poorly understood. To identify these DOM-microbe interactions, we analyzed two continuous 24-m-long sediment cores from the high-energy beach of Spiekeroog Island, Germany. The beach is characterized by a deep subterranean estuary (STE), where sediments are flushed by fresh, saline and brackish water. Even though beach sands are generally low in organic carbon, we found that concentrations of mobilizable DOM from sediments were approximately 16 times higher than the in-situ groundwater concentrations. Ultrahigh-resolution mass spectrometry

☐ <https://link.springer.com/article/10.1007/s10533-026-01315-1>

【环境与气候】

86. High risk from coincidence of extremes in particulate matter and heat in India

☐ Environmental Research Letters | 2026-04-06T23:00:00Z

☐ Abstract:

Exposure to high levels of air pollution and extreme heat stress has been individually linked to significant health and economic problems in India. However, not much is known about the threats resulting from the co-occurrence of these hazards, which have common meteorological drivers. In this study, we perform a country-wide joint risk analysis at the district level, examining the risks of chronic exposure to the joint extremes of particulate matter (PM_{2.5}) pollution (PM_{2.5}, mass of particles smaller than 2.5 μm in diameter) and heat stress (wet bulb globe temperature) in the presence of geophysical and socioeconomic vulnerability. We find that northern, central, and eastern India are at alarming levels of joint risk, potentially exposing 61% (~842 million) of the population and 56% (~1.84

☐ <https://iopscience.iop.org/article/10.1088/1748-9326/ae2346>

【遥感与模型】

87. Modeling saline soil complex permittivity from bound-water microphysics: Laboratory validation at L-/C-bands

☐ Remote Sensing of Environment

☐ Abstract:

Publication date: June 2026Source: Remote Sensing of Environment, Volume 339Author(s): Jundong Wang, Zhigang Sun, Wanxue Zhu, Lidong Ren, Ehsan Eyshi Rezaei

☐ <https://www.sciencedirect.com/science/article/pii/S0034425726001562>

【预印本】

88. Integrated phenomic and transcriptomic analyses unveil superior drought plasticity of North African durum wheat landraces

☐ bioRxiv Plant Biology | 2026-04-07

☐ Abstract:

Drought is a major constraint on the productivity of durum wheat across Mediterranean and North African regions. To elucidate the mechanisms underlying drought resilience, we employed a combination of scenario-controlled phenomics and flag leaf transcriptomics across ten durum wheat genotypes. These included the Tunisian landraces Chili and Mahmoudi, seven breeding lines, and the reference cultivar Svevo. The plants were grown to maturity under well-watered or long-term drought conditions in pots and rhizotrons, enabling a comprehensive assessment of growth, yield components, root architecture, physiological traits, and reaction norm plasticity. Drought markedly reduced performance, yet Chili and Mahmoudi consistently maintained superior biomass, grain number and intrinsic water use effici

☐ <https://doi.org/10.1101/2026.04.03.716342>

【海洋与水生】

89. Optimised thermal-stress indices explain giant clam bleaching observations in a semi-closed atoll lagoon

☐ Coral Reefs | 2026-03-30

☐ Abstract:

Giant clams (*Tridacna maxima*) are abundant in French Polynesia semi-closed atolls, but the sustainability of the ecosystem services they provide has been challenged by recent mass bleaching events. Being able to accurately anticipate the risks of bleaching is needed to implement timely and effective mitigation strategies, but no method currently exists to assess bleaching risk from environmental data in these lagoons. To fill this gap, we used a hind-cast approach to simulate the lagoon temperature experienced by giant clams at Reao Atoll (Tuamotu Archipelago) from 1993 to 2024. We then compared the reconstructed temperature time series with *T. maxima* bleaching states and size structures and densities during three El Niño-induced heatwaves between 2016 and 2024. The 2016 event was the most

☐ <https://link.springer.com/article/10.1007/s00338-026-02860-9>

90. Anthropogenic microparticles ingestion patterns in fish across trophic guilds in mangrove estuaries

☐ Hydrobiologia | 2026-03-30

☐ Abstract:

The ingestion of anthropogenic microparticles by fish is an emerging global concern, with significant knowledge gaps in ecosystems such as mangroves. We analyzed the gastrointestinal tract contents of 736 juvenile fish representing 20 species from different trophic guilds across three estuaries, sampled during two seasons (dry and rainy), within the largest mangrove area in South America. A total of 1516 food items were identified, of which 128 (~ 10%) were classified as anthropogenic particles. Among these particles, 109 were microparticles, 18 were mesoparticles, and one was a macroparticle. Most particles were filaments (84%), predominantly blue and black, across all estuaries and seasons. The occurrence of anthropogenic microparticles averaged 26.07% during the rainy season and 21.25%

☐ <https://link.springer.com/article/10.1007/s10750-026-06151-w>

【预印本】

91. Synchrony Genetics: Linking Ecological Mechanisms to Genetic Structure A framework for genetic inference in ecologically coupled systems

☐ bioRxiv Ecology | 2026-04-04

☐ Abstract:

Spatial synchrony, correlated population dynamics across space, is a defining feature of ecological dynamics, shaping outbreaks, cycles, and waves across ecosystems. Yet its genetic consequences remain poorly resolved because classical population-genetic models assume demographic independence and equilibrium conditions that synchronised populations systematically violate. Here I introduce Synchrony Genetics, a general framework that treats ecological coupling as the causal process and spatial synchrony as only one observable manifestation of that coupling. The framework links the three canonical ecological coupling mechanisms, environmental (Moran-type) coupling, dispersal-driven coupling, interaction-mediated coupling, alone or in combination, to their characteristic genetic signatures. U

☐ <https://doi.org/10.1101/2026.04.02.716123>

92. Disentangling site-specific and shared local adaptation in a classic system of repeated evolution

☐ bioRxiv Evolutionary Biology | 2026-04-07

☐ Abstract:

Repeated divergence across contrasting habitats is widely used to infer natural selection and local adaptation. However, such inferences remain inherently correlative and capture only adaptation shared within habitat types, thereby missing site-specific adaptation among populations from the same habitat type. Field transplant experiments test adaptation more directly by measuring fitness in nature, but they are typically limited to pairwise reciprocal exchanges between populations and therefore cannot separate the contributions of shared habitat-level and site-specific adaptation to fitness. Here, we overcome these limitations by extending the typical transplant framework to include multiple populations transplanted both within and across habitat types. We apply this framework to lake-stre

☐ <https://doi.org/10.1101/2026.04.04.716503>

【植物与农业】

93. Precision planting of cereal rye cover crop improves sweet corn yield and farm benefits

☐ Agronomy for Sustainable Development | 2026-03-17

☐ Abstract:

Including cereal rye (*Secale cereale* L.) in sweet corn (*Zea mays* L.) production systems provides environmental and soil benefits. However, cereal rye often decreases sweet corn yield and thus farm profit. A novel approach to reduce this negative effect is precision planting of cereal rye, which skips the sweet corn row to minimize crop interactions, potentially increasing nitrogen (N) availability, and reducing negative effects of cereal rye on sweet corn. A four-site-yr trial was conducted during the 2022-2024 growing seasons, the first known evaluation of precision-planted cereal rye compared to a no-cover crop control and solid-planted cereal rye. The study evaluated (i) cereal rye morphology, biomass production and its quality, (ii) sweet corn plant population, morphology, and marketab

☐ <https://link.springer.com/article/10.1007/s13593-026-01095-w>

【预印本】

94. Genome-wide characterization of extant clonal diversity in Chilean Carmenere

☐ bioRxiv Genomics | 2026-04-07

☐ Abstract:

Carmenere is a widely cultivated and internationally recognized grapevine cultivar in Chile, yet genetic variation among its clones remains poorly characterized. Early studies based on SSR and AFLP markers detected limited polymorphism, but these approaches interrogate only a small fraction of the genome, leaving the extent of clonal diversity unresolved. Here, we generated an improved chromosome-scale diploid genome assembly of Carmenere FPS clone 02 and characterized clonal genomic diversity by sequencing 36 biological replicates representing 12 clones maintained in Chile, including heritage selections rescued from old producer vineyards by Vina Santa Carolina as part of its Bloque Herencia conservation program, and commercial nursery-derived clones. Focusing on low-frequency variants an

☐ <https://doi.org/10.1101/2026.04.03.716224>

【综合顶刊】

95. Machine and Deep Learning Reveal Sequence Determinants Encoding Bivalent Histone Modifications

☐ Communications Biology | 2026-04-07

☐ Abstract:

Communications Biology, Published online: 07 April 2026; doi:10.1038/s42003-026-09962-8Machine learning and deep learning reveal that bivalent histone modifications in mouse embryonic stem cells is encoded evolutionarily conserved DNA sequences and motifs associated with transcription factors that regulate developmental genes

☐ <https://doi.org/10.1038/s42003-026-09962-8>

96. DNA O-MAP uncovers the molecular neighborhoods associated with specific genomic loci

□ eLife | Tue, 07 Apr 2026 00:

□ Abstract:

The accuracy of crucial nuclear processes such as transcription, replication, and repair depends on the local composition of chromatin and the regulatory proteins that reside there. Understanding these DNA–protein interactions at the level of specific genomic loci has remained challenging due to technical limitations. Here, we introduce a method termed ‘DNA O-MAP’, which uses programmable peroxidase-conjugated oligonucleotide probes to biotinylate nearby proteins. We show that DNA O-MAP can be coupled with label-free or sample multiplexed quantitative proteomics, targeted chemical perturbations, and next-generation sequencing to quantify DNA-proximal proteins and DNA–DNA interactions at specific genomic loci in human and murine cells. Furthermore, we establish that DNA O-MAP is applicable

□ <https://elifesciences.org/articles/102489>

97. Zn-based seed priming enhances drought resistance during germination of *Lespedeza potaninii*: physiological and transcriptomic insights

□ PeerJ | 2026-04-07

□ Abstract:

Lespedeza potaninii is a species of flowering plant in the legume family. Due to its strong stress tolerance and nitrogen-fixing capacity, *Lespedeza potaninii* has become a crucial plant species for reseeded and restoring degraded grasslands in China. However, drought stress during the seedling or germination stage severely impacts the effectiveness of *Lespedeza potaninii*-based grassland restoration efforts. To screen seed-soaking agents enhancing seed germination under sorbitol-simulated drought stress and explore the underlying mechanisms, seed germination, physiological parameters and transcriptomic regulation of *Lespedeza potaninii* seeds under different seed-soaking treatments were analyzed. The results revealed that sorbitol-simulated drought stress significantly reduced seed germination

□ <https://peerj.com/articles/21054>

【生态学核心】

98. Bird occurrence and trophic interactions vary across gradients of tree diversity and microclimate in a planted forest

□ Ecological Applications | Mon, 06 Apr 2026 22:

□ <https://esajournals.onlinelibrary.wiley.com/doi/10.1002/eap.70219>

【环境与气候】

99. Fashion fades, microplastics are eternal: a critical systematic literature review to assess microplastic fibre colours in environmental samples

□ Environmental Research Letters | 2026-04-06T23:00:00Z

☐ Abstract:

Microplastic pollution is a growing environmental concern that requires assessment of a range of parameters. Colour remains a largely overlooked parameter in environmental monitoring, yet a potentially important one from a biological perspective. This systematic review covers 80 peer-reviewed studies published between 2000 and 2020, and addresses microplastic fibres colours across diverse environmental compartments: air, water, sediment and biota. Our analysis reveals that transparent/translucent (T&T), blue, black, and white fibres are the most frequently reported colours globally, with notable variation across environmental compartments and geographical regions. Blue fibres were predominant in biota (32.8%) and water (28.3%), while T&T fibres were prevalent in sediments (35.2%), and yell

☐ <https://iopscience.iop.org/article/10.1088/1748-9326/ae5265>

【植物与农业】

100. Species richness increases vegetation carbon sequestration but decreases soil carbon storage in temperate grasslands of China

☐ Agriculture Ecosystems & Environment

☐ Abstract:

Publication date: 15 August 2026Source: Agriculture, Ecosystems & Environment, Volume 406Author(s): Yingjia Su, Junyu Pu, Jing Huang, Zhixian Xing, Jinshan Du, Yinan Hong, Xinsheng Wang, Yuyang Wang, Kesi Liu

☐ <https://www.sciencedirect.com/science/article/pii/S0167880926002045>

101. Understanding rice yield gaps with crop modeling and machine learning in a long-term continuous cropping experiment

☐ Field Crops Research

☐ Abstract:

Publication date: 15 May 2026Source: Field Crops Research, Volume 342Author(s): Tomoaki Yamaguchi, Ando Mariot Radanielson, Olivyn Angeles, Keisuke Katsura, Kazuki Saito

☐ <https://www.sciencedirect.com/science/article/pii/S037842902600153X>

【土壤与生物地球化学】

102. Divergent chemistry and microbiology, convergent carbon fate: insect frass and plant litter are functionally equivalent under climate warming

☐ Soil Biology and Biochemistry

☐ Abstract:

Publication date: July 2026Source: Soil Biology and Biochemistry, Volume 218Author(s): Ying Liu, Jiaqi Wu, Yifan Wang, Jiaxin Yang, Haiyang Zhang, Liya Wei, Qinglin Yin, Yidi Wang, Xinyu Wang, Liangchao Jiang, Yong Jiang, Xingguo Han, Jing Wang

☐ <https://www.sciencedirect.com/science/article/pii/S0038071726000787>

【预印本】

103. An agent-based approach for designing effective protection

☐ bioRxiv Ecology | 2026-04-07

☐ Abstract:

The main threat to New Zealand dolphins is entanglement in fishing nets. We developed an agent-based model (ABM) to assess how area-based controls on fishing methods can reduce fishing mortality and population declines. The model incorporates the behavior and distributions of dolphins and fishing vessels, and realistic displacement of fishing effort when protection is extended. It was designed and calibrated using pattern-oriented modeling. Our results show that mortality due to entanglement in fishing gears has been reduced thanks to a gradual increase in dolphin protection. However, current protection is not as effective as previously thought, and scarce populations are negatively affected by Allee effects. Neither national nor international goals for reducing bycatch are met by current

☐ <https://doi.org/10.1101/2026.04.03.716393>

104. Sharing the trail: recreation effects on bear behaviour in a Canadian Rocky Mountain Park

☐ bioRxiv Ecology | 2026-04-04

☐ Abstract:

Humans are profoundly reshaping the natural world. These changes are giving rise to complex and mutually risky dynamics between people and large carnivores. In protected areas across North America, bears (*Ursus* sp.) face rapidly rising recreation pressures that can alter their use of the landscape, either displacing them from high-quality habitats or drawing them into human-wildlife conflicts through habituation or attraction to anthropogenic resources. However, disentangling responses to recreation from other drivers can be difficult because human activity covaries with environmental and seasonal processes that also shape bear activity. We leveraged the partial closure of the popular Berg Lake Trail in Mount Robson Park, British Columbia, Canada, to investigate whether black (*Ursus* *americanus*)

☐ <https://doi.org/10.1101/2026.04.02.714576>

【综合顶刊】

105. Investigating the native functions of [NiFe]-CODH through genomic context analysis

☐ eLife | Tue, 07 Apr 2026 00:

☐ Abstract:

Carbon monoxide dehydrogenases containing nickel-iron active sites ([NiFe]-CODHs) catalyze the reversible oxidation of CO to CO₂, representing key targets for biocatalytic CO₂ reduction. Despite dramatic differences in catalytic rates and O₂ tolerance between CODH variants, the molecular basis for this functional diversity remains poorly understood. We applied comparative genomics and synteny analysis to investigate the biochemical roles of CODH clades A-F using 1376 CODH and 1545 hybrid cluster protein sequences. Around 30% of genomes encode multiple CODH isoforms. Analysis revealed distinct gene clustering patterns correlating with biochemical function. Clades A, E, and F exhibit a degree of distributional exclusivity. Clades C and D frequently co-occur with active CODHs, suggesting auxi

☐ <https://elifesciences.org/articles/108780>

【植物与农业】

106. Re-Engineering Resilience: Predictive Pheno-Metabolomics and Machine Learning for Climate-Adaptive Crop Breeding

☐ Plant Cell & Environment | Mon, 06 Apr 2026 07:

☐ Abstract:

Plant, Cell & Environment, Volume 49, Issue 5, Page 2387-2389, May 2026.

☐ <https://onlinelibrary.wiley.com/doi/10.1111/pce.70397>

【遥感与模型】

107. Explainable AI -driven deep learning model for the root image-based classification of the medicinal plants *Saposhnikovia divaricata*, *Glehnia littoralis*, and *Peucedanum japonicum*

☐ Ecological Informatics

☐ Abstract:

Publication date: May 2026
Source: Ecological Informatics, Volume 95
Author(s): Hyein Lee, Yu-Jin Jeon, So Jin Park, Ho-Youn Kim, Dahye Ryu, Jung-Ok Kim, Joo-Young Kim, Dae-Hyun Jung

☐ <https://www.sciencedirect.com/science/article/pii/S1574954126001494>

【预印本】

108. Endogenous mutational mechanisms and metabolic context shape endometrial cancer

☐ bioRxiv Genomics | 2026-04-07

☐ Abstract:

Uterine corpus endometrial carcinoma (UCEC) is a common gynecologic malignancy with rising mortality, yet its genome-wide mutational architecture remains incompletely understood. Here we analyze deep whole-genome sequencing data from 440 UCEC tumors from The Cancer Genome Atlas, integrated with transcriptomic and clinical data, to define subtype-specific mutational processes and genomic architectures. We uncover pronounced molecular subtype-specific differences in endogenous mutational mechanisms, retrotransposition activity, structural variation, and chromosomal instability. LINE-1 retrotransposition emerges as a key contributor to genome instability in UCEC, acting as a prominent source of structural variation in copy-number stable tumors and associating with chromothripsis and ecDNA-med

☐ <https://doi.org/10.1101/2026.04.03.716356>

【生态学核心】

109. Designing networks of freshwater and marine conservation areas accounting for cross-realm connectivity and threat propagation

☐ Journal of Applied Ecology | Mon, 06 Apr 2026 22:

☐ <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2664.70347>

110. Land Use Influences on Freshwater Biodiversity

☐ Global Ecology and Biogeography | Wed, 01 Apr 2026 04:

☐ <https://onlinelibrary.wiley.com/doi/10.1111/geb.70230>

111. The persistent advantage of model-based phylogenetic methods for single-trait prediction

☐ Methods in Ecology and Evolution | Tue, 31 Mar 2026 21:

☐ Abstract:

Methods in Ecology and Evolution, Volume 17, Issue 4, Page 1032-1041, April 2026.

☐ <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210x.70258>

【海洋与水生】

112. Zooplankton diel vertical migration enhances carbon export via distinct mechanisms in a warming North Pacific

☐ Limnology and Oceanography Letters | Sun, 05 Apr 2026 21:

☐ <https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lol2.70124>

113. Effects-based assessment of water and benthic habitat quality using coral larvae

☐ Coral Reefs | 2026-03-29

☐ Abstract:

Larval performance is a critical bottleneck in coral restoration using in situ seeding approaches, yet early-stage larval survivorship and settlement remain difficult to predict based on site appearance alone. Here, we applied an effects-based methods (EBM) approach using *Diploria labyrinthiformis* larvae to evaluate whether site-specific environmental conditions influenced larval performance at a previously unsuccessful in situ culture site in Bonaire. Larvae were exposed to seawater collected from seven locations, including stratified depths and a reference site, and to settlement substrates (seeding units) pre-conditioned at four different locations. Larval survivorship was significantly lower in surface water from the original site compared to all other treatments, while survivorship in

☐ <https://link.springer.com/article/10.1007/s00338-026-02855-6>

【保护与生物多样性】

114. Incorporating Biotic Interactions and Dispersal Processes in Modelling Species Distributions of Freshwater Organisms: A Review

☐ Diversity and Distributions | Wed, 01 Apr 2026 00:

☐ <https://onlinelibrary.wiley.com/doi/10.1111/ddi.70163>

【土壤与生物地球化学】

115. Spurious nitrous oxide emissions from acidic nitrite solutions: a cautionary tale on the critical role of experimental apparatus

☐ Biogeochemistry | 2026-03-26

☐ Abstract:

Nitrite (NO_2^-) is known to undergo disproportionation into nitric oxide (NO) and nitrogen dioxide (NO_2) via an abiotic pathway in acidic aqueous solutions. However, whether a direct pathway for its conversion to nitrous oxide (N_2O) exists remains unclear. This study investigated this potential pathway under both ambient air and nitrogen-purged conditions. Initial experiments yielded substantial N_2O production, appearing to validate a novel abiotic process in the nitrogen cycle. However, through a series of exclusion experiments involving physical barriers and comparative analysis of sealing materials, these emissions were revealed to be entirely spurious. Analysis of water extracts from these materials demonstrated dissolved organic carbon (DOC) concentrations ranging from 37.67 to 736.8 μ

☐ <https://link.springer.com/article/10.1007/s10533-026-01319-x>

116. Enhanced near-surface soil labile carbon and microbial indicators under cover crops and crop diversification in long-term no-till systems

☐ Geoderma

☐ Abstract:

Publication date: May 2026
Source: Geoderma, Volume 469
Author(s): Sainfort Vital, Sutie Xu, Namrata Ghimire, Peter Sexton, Jixiang Wu, Christopher Graham, Lan Xu, Mengistu Geza

☐ <https://www.sciencedirect.com/science/article/pii/S0016706126001369>

【遥感与模型】

117. Vertical distribution of type-discriminated aerosol concentration from a three-wavelength backscatter spaceborne lidar

☐ Remote Sensing of Environment

☐ Abstract:

Publication date: June 2026 Source: Remote Sensing of Environment, Volume 339 Author(s): Fazzal Qayyum, Juan Cuesta, Abou Bakr Merdji, Anton Lopatin, Oleg Dubovik, Durgesh Nandan Piyush, Laaziz El Amraoui

☐ <https://www.sciencedirect.com/science/article/pii/S0034425726001690>

【生态学核心】

118. Aridity amplifies soil microbial influences and dampens plant diversity effects on lakeshore wetland multifunctionality

☐ Journal of Ecology | Wed, 01 Apr 2026 22:

☐ <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2745.70301>

119. Increases in species richness cause decreases in phylogenetic diversity in Mediterranean and Temperate plant assemblages

☐ Ecology | Wed, 01 Apr 2026 05:

☐ <https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecy.70361>

120. Realm Specific Adaptations Lead to Contrasting Global Patterns in Submerged and Emergent Aquatic Plant Height

☐ Global Ecology and Biogeography | Sun, 05 Apr 2026 21:

☐ <https://onlinelibrary.wiley.com/doi/10.1111/geb.70232>

121. Necromass carbon but not microbes constrain soil carbon release in restoration of degraded alpine grassland

☐ Ecological Applications | Mon, 06 Apr 2026 22:

☐ <https://esajournals.onlinelibrary.wiley.com/doi/10.1002/eap.70225>

122. Historical data reveal extirpation of foundation species and kelp forest community deborealization in a coastal hotspot

☐ Ecological Applications | Mon, 06 Apr 2026 19:

☐ <https://esajournals.onlinelibrary.wiley.com/doi/10.1002/eap.70223>

【海洋与水生】

123. Connected reservoirs: Modeling aquatic ecosystems along a cascade system in Brazil

☐ Limnology and Oceanography Letters | Sun, 05 Apr 2026 21:

☐ <https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lol2.70122>

124. Small jellyfish, large consequences: The overlooked predatory role of hydromedusae in subtropical estuarine ecosystems

☐ Limnology and Oceanography | Sun, 05 Apr 2026 21:

☐ <https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lno.70360>

【保护与生物多样性】

125. Addressing Gaps in Butterfly Population Monitoring to Catalyze Global Insect Conservation

☐ Conservation Letters | Mon, 23 Mar 2026 00:

☐ Abstract:

Conservation Letters, Volume 19, Issue 2, March/April 2026.

☐ <https://conbio.onlinelibrary.wiley.com/doi/10.1111/con4.70037>

126. Prey Partitioning in a Diverse Carnivore Community: Implications for Reintroduced Fishers in Washington

☐ Animal Conservation | Tue, 17 Mar 2026 04:

☐ Abstract:

Animal Conservation, EarlyView.

☐ <https://zslpublications.onlinelibrary.wiley.com/doi/10.1111/acv.70062>

【植物与农业】

127. Effects of nitrogen fertilization on forage production, animal performance, and greenhouse gas emissions in tropical pastures: A systematic review

☐ Agriculture Ecosystems & Environment

☐ Abstract:

Publication date: 15 August 2026
Source: Agriculture, Ecosystems & Environment, Volume 406
Author(s): Hitalo Rodrigues da Silva, Gelson dos Santos Difante, Francisca Fernanda da Silva Roberto, Marcos Antônio Ferreira Júnior, Carolina Marques Costa Araújo, Vanessa Zironi Longhini, Jéssica Gomes Rodrigues, Marislayne de Gusmão Pereira, Denise Baptaglin Montagner, Luís Carlos Vinhas Ítavo, Antônio Leandro Chaves Gurgel, Marina Maria Pedrosa Méca Ferreira de Castro

☐ <https://www.sciencedirect.com/science/article/pii/S016788092600188X>

128. Physiology and Transcriptomics Reveal Divergent Strategies of Mycorrhiza-Mediated Drought Adaptation in Poplar

☐ Plant Cell & Environment | Mon, 06 Apr 2026 22:

☐ Abstract:

Plant, Cell & Environment, EarlyView.

☐ <https://onlinelibrary.wiley.com/doi/10.1111/pce.70511>

129. Silene, a versatile model system: from sex and genome evolution to ecology and speciation

☐ New Phytologist | Sun, 05 Apr 2026 22:

☐ Abstract:

New Phytologist, EarlyView.

☐ <https://nph.onlinelibrary.wiley.com/doi/10.1111/nph.71153>

130. A conceptual framework for simultaneous optimization of integrated climate scenario data and phenology models

☐ Agronomy for Sustainable Development | 2026-03-12

☐ Abstract:

Climate-resilient crop varieties are exceedingly important for global food security. High-accuracy predictions of crop phenology in future climate scenarios require a more precise description of putative crop-environment interactions. While the accuracy of these predictions is affected by three primary sources of uncertainty —the phenology model, the climate scenario data, and their interaction —commonly used approaches consider only the phenology model as a source of uncertainty, while uncertainties arising from climate scenario data and their interaction with the model are commonly neglected. To address this gap, we developed a novel dynamic crop phenology model framework that simultaneously integrates uncertainties arising from all three sources. This model framework was developed using

☐ <https://link.springer.com/article/10.1007/s13593-026-01091-0>

【预印本】

131. Semen collection, short term storage, and cryopreservation in the Texas horned lizard (*Phrynosoma cornutum*)

☐ bioRxiv Zoology | 2026-04-06

☐ Abstract:

As global reptile populations continue to decline, improving reproductive success in managed populations of listed species, such as *Phrynosoma cornutum* (the Texas horned lizard) has become increasingly critical for species survival. One understudied area of reproductive research in reptile species is gamete collection and storage, a crucial component for maintaining genetic diversity. In Texas, semen was collected from wild *P. cornutum* (n = 20) in June 2025. Semen collection was performed via electroejaculation (EEJ) under alfaxalone anesthesia. Prior to semen collection, snout-vent-lengths (SVL) and weights were recorded and testes measurements were taken using a portable ultrasound. Average sperm motility and concentration across all lizards was 83.7% and 85.7 x 10⁶ sperm/mL, respectively.

☐ <https://doi.org/10.1101/2026.04.03.716302>

【遥感与模型】

132. A first characterization of lake ice thickness on the Tibetan Plateau by leveraging satellite altimetry and ERA5 reanalysis

☐ International Journal of Applied Earth Observation

☐ Abstract:

Publication date: May 2026
Source: International Journal of Applied Earth Observation and Geoinformation, Volume 149
Author(s): Tian Xia, Laihui Luo, Wenbin Zhu, Jianting Ju, Liguang Jiang

☐ <https://www.sciencedirect.com/science/article/pii/S1569843226002001>

133. NatShore: Automated nation-scale shoreline extraction and change analysis using multi-decadal cross-mission remote sensing data

☐ International Journal of Applied Earth Observation

☐ Abstract:

Publication date: May 2026
Source: International Journal of Applied Earth Observation and Geoinformation, Volume 149
Author(s): Ho Tin Hung, Ya-Lun S. Tsai

☐ <https://www.sciencedirect.com/science/article/pii/S1569843226001937>

【综合顶刊】

134. Metformin attenuates TBHP-induced oxidative injury in human lens epithelial cells and is associated with SIRT1/FOXO1-related autophagy

☐ PLOS ONE | 2026-04-07T14:00:00Z

☐ Abstract:

by Zhuxuan Yan, Wei Chi, Zhenguo Yan, Hanrui Wang Metformin (MET), a first-line antidiabetic drug, has been increasingly implicated in cellular protection under oxidative stress, yet its mechanisms in lens epithelial cells (LECs) remain incompletely defined. Using a tert-butyl hydroperoxide (TBHP)-induced acute oxidative injury model in human HLE-B3 cells, we investigated whether SIRT1/FOXO1-related autophagy contributes to MET-associated cytoprotection. MET pretreatment reduced intracellular reactive oxygen species, preserved antioxidant defenses, improved cell viability, and decreased apoptosis after TBHP challenge. MET also enhanced autophagy markers and, under lysosomal blockade with chloroquine or bafilomycin A1, showed LC3-II/p62 changes consistent with increased autophagic flux. Pha

☐ <https://journals.plos.org/plosone/article>

135. State impulsivity and substance use: A systematic review and meta-analysis protocol

☐ PLOS ONE | 2026-04-07T14:00:00Z

☐ Abstract:

by Ashmita Mazumder, Suzanne Erb, Marc A. Fournier The association between substance use and impulsivity has been documented extensively in the literature. More recently, there has been a shift from viewing impulsivity solely as a stable trait toward examining its moment-to-moment expression in everyday life and how these fluctuations influence substance use. Despite this growing interest, there has not yet been a comprehensive meta-analysis synthesizing findings across studies. Existing reviews have largely focused on trait-level impulsivity, which limits our understanding of how impulsivity operates across real-world contexts. The aim of this proposed meta-analysis is to integrate these findings and quantify the strength of the association between everyday impulsivity and substance use,

☐ <https://journals.plos.org/plosone/article>

【生态学核心】

136. Root exudates mediate microorganisms to improve rhizosphere nutrient availability and crop productivity under straw mulching and slow-release nitrogen fertilizer

☐ Functional Ecology | Sun, 05 Apr 2026 23:

☐ Abstract:

Functional Ecology, EarlyView.

☐ <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2435.70327>

137. Mycorrhizal symbiosis and environmental conditions shape understory herb diversity in a large temperate forest region

☐ Functional Ecology | Sat, 04 Apr 2026 00:

☐ Abstract:

Functional Ecology, EarlyView.

☐ <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2435.70323>

138. Forest type and leaf habit mediate thermal and drought tolerance across a tropical elevational gradient

☐ Functional Ecology | Sat, 04 Apr 2026 00:

☐ Abstract:

Functional Ecology, EarlyView.

☐ <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2435.70328>

139. Habitat heterogeneity and anthropogenic disturbance jointly affect species spatial associations and persistence

☐ Ecological Monographs | Sat, 21 Mar 2026 02:

☐ <https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecm.70059>

140. Catchment conversion to agriculture alters freshwater macroinvertebrate community responses to flow disturbance: results from a replicated in-stream experiment

☐ Oikos | Fri, 03 Apr 2026 06:

☐ Abstract:

Oikos, EarlyView.

☐ <https://nsojournals.onlinelibrary.wiley.com/doi/10.1002/oik.12033>

141. adapt3: Adaptive dynamics and community projection in the R programming language

☐ Methods in Ecology and Evolution | Mon, 06 Apr 2026 05:

☐ Abstract:

Methods in Ecology and Evolution, EarlyView.

☐ <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210x.70298>

142. fastbioclim: An R package for creating custom-time bioclimatic and derived environmental summary variables

☐ Methods in Ecology and Evolution | Wed, 01 Apr 2026 05:

☐ Abstract:

Methods in Ecology and Evolution, EarlyView.

☐ <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/2041-210x.70291>

【综合顶刊】

143. Safeguarding macro-financial stability under carbon pricing and rapid energy transition

☐ Communications Earth & Environment | 2026-04-07

☐ Abstract:

Communications Earth & Environment, Published online: 07 April 2026; doi:10.1038/s43247-026-03209-4 Rapid decarbonization driven by carbon pricing can pose macro-financial stability risks, which targeted fiscal and monetary policies can help mitigate, according to integrated assessment and agent-based modeling frameworks.

☐ <https://doi.org/10.1038/s43247-026-03209-4>

【环境与气候】

144. Integrating climate adaptation and peacebuilding: capacity development in climate and conflict-affected communities

☐ Global Environmental Change

☐ Abstract:

Publication date: July 2026 Source: Global Environmental Change, Volume 98 Author(s): Luisa Fernanda Bedoya Taborda, Tiffany H. Morrison, Michele L. Barnes

☐ <https://www.sciencedirect.com/science/article/pii/S0959378026000403>

145. Land titles and deforestation: evidence from Peru's oil palm sector

☐ Global Environmental Change

☐ Abstract:

Publication date: July 2026 Source: Global Environmental Change, Volume 98 Author(s): Celeste Gunderson, Jess L'Roe, Lisa Rausch, Clare Sullivan, Holly Gibbs, Lisa Naughton-Treves

☐ <https://www.sciencedirect.com/science/article/pii/S0959378026000415>

146. Metabolic engineering of *Halomonas cupida* for co-mineralization of phenol and p-nitrophenol in high-saline wastewater

☐ Journal of Environmental Management

☐ Abstract:

Publication date: 15 April 2026 Source: Journal of Environmental Management, Volume 404 Author(s): Ziling Yuan, Yujie Liu, Haomin Chen, Jing Dang, Xinyu Zhang, Xinyao Li, Ruihua Liu, Chao Yang

☐ <https://www.sciencedirect.com/science/article/pii/S0301479726010571>

147. Sex hormone disruption induced by biodegradable plastic extracts compared with conventional plastics in H295R cells and adult male zebrafish

☐ Environmental Pollution

☐ Abstract:

Publication date: 1 June 2026 Source: Environmental Pollution, Volume 398 Author(s): Eunhye Kim, Aulia Salmad-diina, Ah-Reum Jo, Mike Williams, Kyungho Choi

☐ <https://www.sciencedirect.com/science/article/pii/S0269749126003878>

148. Improved health risk management of heavy metals in cigarette smoke from middle Chinese best-selling cigarettes using respiratory tract deposition modeling and simulated lung fluid extraction

☐ Environmental Pollution

☐ Abstract:

Publication date: 1 June 2026 Source: Environmental Pollution, Volume 398 Author(s): Jinyuan Guo, Wenzhuo Deng, Mian Wu, Fei Li, Ruizhi Zhang

☐ <https://www.sciencedirect.com/science/article/pii/S0269749126004380>

【海洋与水生】

149. Shifting Trends in Aquaculture's Biological Potential to Address Food, Climate and Biodiversity Challenges

☐ Fish and Fisheries | Wed, 25 Mar 2026 12:

☐ Abstract:

Fish and Fisheries, EarlyView.

☐ <https://onlinelibrary.wiley.com/doi/10.1111/faf.70081>

150. Mesozooplankton trophic structure in the Argo Basin, northwestern Australia

☐ Deep Sea Research II

☐ Abstract:

Publication date: June 2026 Source: Deep Sea Research Part II: Topical Studies in Oceanography, Volume 227 Author(s): Lindsey E.-H. Kim, Rasmus Swalethorp, Moira Décima, Grace F. Cawley, Michael R. Landry

☐ <https://www.sciencedirect.com/science/article/pii/S096706452600041X>

【保护与生物多样性】

151. Pleistocene Divergence and Long-Term Population Decline in Three Endemic Euphorbia Species of High Conservation Concern From the South-Western Alps and the Mountains of Corsica and Sardinia

☐ Diversity and Distributions | Tue, 31 Mar 2026 00:

☐ <https://onlinelibrary.wiley.com/doi/10.1111/ddi.70177>

152. Better research is required to clarify the genetic composition of the Apennine Italian wolf population

☐ Biological Conservation

☐ Abstract:

Publication date: June 2026 Source: Biological Conservation, Volume 318 Author(s): Ettore Randi, Elena Fabbri, Marco Galaverni, Romolo Caniglia

☐ <https://www.sciencedirect.com/science/article/pii/S0006320726001448>

153. Restoring surf and turf: Transferable lessons between marine and terrestrial restoration

☐ Biological Conservation

☐ Abstract:

Publication date: June 2026 Source: Biological Conservation, Volume 318 Author(s): Trevyn A. Toone, Emilee D. Benjamin, Julie Larson, Mary-Margaret S. McKinney, James C. Russell, Carter Smith, Y. Stacy Zhang

☐ <https://www.sciencedirect.com/science/article/pii/S0006320726001552>

154. Introduced species are a global threat to lizard biodiversity

☐ Biological Conservation

☐ Abstract:

Publication date: June 2026Source: Biological Conservation, Volume 318Author(s): Nicole Stevens, Lee Fitzgerald

☐ <https://www.sciencedirect.com/science/article/pii/S0006320726001400>

155. Genome-Wide DNA Methylation Patterns Predict Age in the Zebra Shark (*Stegostoma tigrinum*) and Provide Insight Into the Evolution of Vertebrate Aging

☐ Molecular Ecology | Fri, 03 Apr 2026 04:

☐ <https://onlinelibrary.wiley.com/doi/10.1111/mec.70326>

156. Global Biodiversity Framework Targets Can Drive Action on Insect Declines, but Lack Robust Indicators to Prove Their Effectiveness

☐ Conservation Letters | Wed, 18 Mar 2026 11:

☐ Abstract:

Conservation Letters, Volume 19, Issue 2, March/April 2026.

☐ <https://conbio.onlinelibrary.wiley.com/doi/10.1111/con4.70025>

【植物与农业】

157. Crop diversification drives microbial-mediated nitrogen accumulation via fungal necromass stabilization

☐ Agriculture Ecosystems & Environment

☐ Abstract:

Publication date: 15 August 2026Source: Agriculture, Ecosystems & Environment, Volume 406Author(s): Chunyan Liu, Guixin Zhang, Wentao Zhang, Yingxin Lu, Haishui Yang, Feng-Min Li, Jie Zhou, Lingling Shi

☐ <https://www.sciencedirect.com/science/article/pii/S0167880926002057>

158. Trade-offs between crop yield and soil organic carbon sequestration under straw return across cropping systems

☐ Field Crops Research

☐ Abstract:

Publication date: 15 May 2026Source: Field Crops Research, Volume 342Author(s): Yufei Cui, Mengdi Yang, Xiao Yang, Rui Wang, Shengli Guo

☐ <https://www.sciencedirect.com/science/article/pii/S0378429026001528>

【土壤与生物地球化学】

159. From soil to gas –high resolution insights into plant-soil interactions by integrating planar oxygen optodes, porewater chemistry, soil microbial analysis and trace soil gas flux using a rhizobox approach

☐ Soil Biology and Biochemistry

☐ Abstract:

Publication date: July 2026Source: Soil Biology and Biochemistry, Volume 218Author(s): Sean Fettrow, Pauline Sophie Rummel, William James, Michael J. Wilkins, Klaus Koren, Thomas Borch

☐ <https://www.sciencedirect.com/science/article/pii/S0038071726000684>

160. Global Natural Soil N₂O Emissions: An Analysis With a Global Biogeochemistry Model From 1990 to 2023

☐ Global Biogeochemical Cycles | Sat, 04 Apr 2026 03:

☐ <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025GB008804>

161. Empirical evaluation and recommendations for equivalent soil mass adjustment of soil organic carbon stocks

☐ Geoderma

☐ Abstract:

Publication date: May 2026Source: Geoderma, Volume 469Author(s): Eric Potash, Adam C. von Haden, Luke Bergschneider, Michael S. Douglass, Lenarth Ferrari, Rosa Ibarra Lopez, Adriana Reconco Martinez, Andrew J. Margenot

☐ <https://www.sciencedirect.com/science/article/pii/S0016706126001291>

【遥感与模型】

162. Irrigation buffers climate influence: Stable parameter sensitivities in APSIM-wheat under warm-temperate monsoon conditions

☐ Ecological Modelling

☐ Abstract:

Publication date: July 2026Source: Ecological Modelling, Volume 517Author(s): Hongqin Wang, Lei Li, Jiaxue Li, Jinbiao Xiang, Yong He

☐ <https://www.sciencedirect.com/science/article/pii/S0304380026001158>

163. A new framework for measuring community diversity in temporal ecological networks

☐ Ecological Informatics

☐ Abstract:

Publication date: May 2026
Source: Ecological Informatics, Volume 95
Author(s): Zakariya Ghalmane, Pascal Poncellet, Roberto Interdonado, Dino Ienco, Pierluigi Carbonara, Angelo Cau, Antonio Esteban, Maria Teresa Farriols, Maria Cristina Follesa, Cristina Garcia-Ruiz, Germana Garofalo, Gioacchino Bono, Taha Imzilen, Igor Isajlović, Stefanos Kavadas, Irida Maina, Porzia Maiorano, Chiara Manfredi, Jurgen Mifsud, Panagiota Peristeraki

☐ <https://www.sciencedirect.com/science/article/pii/S1574954126001226>

【预印本】

164. Ancient Ryukyu Jomon contributed to past and current genetic structure of Japanese populations

☐ bioRxiv Evolutionary Biology | 2026-04-07

☐ Abstract:

Characterized by the earliest use of pottery, the Jomon culture was a unique Neolithic culture that spread throughout the Japanese Archipelago. Previous archaeological evidence suggests that Jomon hunter-gatherers colonized the southernmost islands, the Ryukyu Archipelago, by approximately 7,000 years before present (YBP). However, genetic characteristics of the Ryukyu Jomon population and its contribution to the modern population have not been elucidated yet. In this study, we newly sequenced 273 modern and 25 ancient (6,700-900 YBP) whole genomes collected across the Ryukyu Archipelago. Our analysis demonstrated a genetic differentiation between the Hondo (Japanese mainland) and Ryukyu Jomon, dating back to ~6,900 YBP. After the divergence from the Hondo Jomon, the Ryukyu Jomon experienc

☐ <https://doi.org/10.1101/2026.04.03.712818>

165. Forensic Identification of Confiscated Helmeted Hornbill (*Rhinoplax vigil*) Casques and Implications for Individual Quantification in Wildlife Crimes

☐ bioRxiv Zoology | 2026-04-06

☐ Abstract:

In wildlife forensic practice, species identification and estimation of the Minimum Number of Individuals (MNI) for highly processed specimens have long relied on weight-based conversion methods, which may result in underestimation of the number of individuals involved in a case. Focusing on confiscated casque products of the helmeted hornbill (*Rhinoplax vigil*), this study combines macroscopic morphological examination with mitochondrial DNA barcoding (16S rRNA, COI, and Cytb) to explore a more robust approach for individual quantification. The results demonstrate that the conventional "weight-based" approach overlooks critical biological information contained in anatomical structures and cannot accurately reflect the actual number of individuals involved. Based on this, we propose an anat

☐ <https://doi.org/10.1101/2026.04.02.715475>

166. Investigating cognitive enrichment for dairy calves through behavioral measures of participation and engagement: a pilot study

☐ bioRxiv Zoology | 2026-04-04

☐ Abstract:

Introducing cognitive enrichment from an early age has the potential to enhance an animals capacity to learn both simple and complex tasks, promote neural plasticity, and support cognitive development. This is applicable for young cattle who are at a critical stage in their development and could benefit from the influence cognitive enrichment has on their behavioral expression. This study aims to explore the effects cognitive enrichment has on weaned dairy calves through analyzing behavioral measures of voluntary participation and short-term behavioral reactions to enrichment exposure. Our study involved a total of five pairs of weaned calves (n=8 treatment; n=2 control). The treatment groups were presented with three variations of a puzzle box, each equipped with unique challenges that of

☐ <https://doi.org/10.1101/2026.04.01.715895>

【综合顶刊】

167. Torus-margo pits cannot function in vessel-bearing angiosperms

☐ Communications Biology | 2026-04-06

☐ Abstract:

Communications Biology, Published online: 06 April 2026; doi:10.1038/s42003-026-09800-xMechanistic modelling identifies conduit geometric constraints on torus-margo membrane function, resolving the surprising absence of this low-resistance, high-safety design in vessel-bearing plants.

☐ <https://doi.org/10.1038/s42003-026-09800-x>

【生态学核心】

168. Structural diversity surpasses species diversity in regulating the spatial variation of grass-land productivity

☐ Journal of Ecology | Mon, 06 Apr 2026 00:

☐ <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2745.70294>

【综合顶刊】

169. Urinary incontinence management in Chinese primary health institutions: findings from a regional survey

☐ PeerJ | 2026-04-07

☐ Abstract:

Background Urinary incontinence (UI) is a prevalent yet under-recognized condition among women in China, impairing quality of life and imposing socioeconomic burden. Primary care providers are pivotal for early detection and first-line management. Unlike Western systems dominated by general practitioners, China's primary women's health care is largely provided by gynecologists in county-level or lower-tier institutions. Evidence on their UI-related knowledge and practices is limited, hampering targeted training and policy efforts. Methods We conducted a cross-sectional survey (March 2023 – September 2024) among gynecologists in Fujian primary healthcare institutions recruited via professional WeChat groups. A structured, expert-developed questionnaire assessed demographics, UI-related aware

☐ <https://peerj.com/articles/21079>

【生态学核心】

170. Coming out of the cold: Vanadium is an important player in N-fixation in warm tropical environments

☐ Ecology | Wed, 01 Apr 2026 05:

☐ <https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecy.70367>

171. Vegetation succession triggers a non-linear trajectory of SOC molecular architecture driven by microbial resource acquisition strategies

☐ Journal of Applied Ecology | Tue, 07 Apr 2026 00:

☐ <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2664.70356>

172. Rapid ecological and evolutionary divergence during a poleward range expansion

☐ Ecological Monographs | Sat, 28 Mar 2026 03:

☐ <https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecm.70047>

173. Using recommender systems to analyze and predict plant communities from partial assemblages

☐ Ecological Monographs | Mon, 23 Mar 2026 20:

☐ <https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecm.70056>

174. Biogeography and residence time influence the climatic niche shift of invasive plants in India

☐ Oikos | Fri, 03 Apr 2026 06:

☐ Abstract:

Oikos, EarlyView.

☐ <https://nsojournals.onlinelibrary.wiley.com/doi/10.1002/oik.11611>

175. The acoustic niche hypothesis meets habitat fragmentation and isolation

☐ Journal of Animal Ecology | Tue, 31 Mar 2026 22:

☐ Abstract:

Journal of Animal Ecology, Volume 95, Issue 4, Page 574-576, April 2026.

☐ <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2656.70206>

176. Spatial distribution of acoustic traits in bird assemblages along regional bioclimatic gradients

☐ Ecography | Fri, 03 Apr 2026 06:

☐ Abstract:

Ecography, EarlyView.

☐ <https://nsojournals.onlinelibrary.wiley.com/doi/10.1002/ecog.08011>

【环境与气候】

177. [ASAP] Alpine Wetlands as an Overlooked Source of Dimethyl Sulfide (DMS) on the Tibetan Plateau

☐ ES&T Letters | Tue, 31 Mar 2026 02:

☐ Abstract:

Environmental Science & Technology Letters DOI: 10.1021/acs.estlett.6c00135

☐ <https://doi.org/10.1021/acs.estlett.6c00135>

178. How are waste management technologies structurally organized and evolving over time?

☐ Journal of Environmental Management

☐ Abstract:

Publication date: 15 April 2026 Source: Journal of Environmental Management, Volume 404 Author(s): Mohammad Sabet, Navid Mohammadi

☐ <https://www.sciencedirect.com/science/article/pii/S0301479726010601>

179. Associations of urinary and follicular fluid phthalate exposure with salpingitis in women seeking fertility treatment

☐ Environmental Pollution

☐ Abstract:

Publication date: 1 June 2026 Source: Environmental Pollution, Volume 398 Author(s): Qingqing Hou, Hongmei Liao, Zhimo Zhang, Jiaying Yuan, Qingchun Guo, Yi Wang, Juan Li, Yaoyao Du, Qiang Zeng, Taoran Deng, Yufeng Li

☐ <https://www.sciencedirect.com/science/article/pii/S0269749126004276>

【海洋与水生】

180. Temporal dynamics of mesopelagic fishes within a mesoscale eddy: A Lagrangian perspective

☐ Limnology and Oceanography Letters | Mon, 30 Mar 2026 00:

☐ <https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lol2.70113>

181. Water physical and chemical properties override urbanization in shaping diatom taxonomic and functional diversity in urban ponds

☐ Limnology and Oceanography | Fri, 03 Apr 2026 21:

☐ <https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lno.70363>

182. Bite, swab, identify: Validating molecular tools for detecting depredating shark species

☐ Journal of Fish Biology | Mon, 06 Apr 2026 08:

☐ Abstract:

Journal of Fish Biology, Volume 108, Issue 2, Page 778-781, February 2026.

☐ <https://onlinelibrary.wiley.com/doi/10.1111/jfb.70264>

183. The SPoRC Stock Assessment Package: A Generalized Next-Generation Platform to Assess Spatial, Age and Sex-Structured Populations

☐ Fish and Fisheries | Mon, 30 Mar 2026 00:

☐ Abstract:

Fish and Fisheries, EarlyView.

☐ <https://onlinelibrary.wiley.com/doi/10.1111/faf.70082>

184. Free Tools—Not So “Free”? The Evolution, Risks and Costs of Fishery Stock Assessment Tools

☐ Fish and Fisheries | Wed, 25 Mar 2026 12:

☐ Abstract:

Fish and Fisheries, EarlyView.

☐ <https://onlinelibrary.wiley.com/doi/10.1111/faf.70076>

185. Interannual variability in the Sulu Sea heat content and the deep-reaching warming trend

☐ Deep Sea Research II

☐ Abstract:

Publication date: June 2026Source: Deep Sea Research Part II: Topical Studies in Oceanography, Volume 227Author(s): Yuchun Jin, Yuanlong Li, Yuanyuan Song, Fan Wang

☐ <https://www.sciencedirect.com/science/article/pii/S0967064526000524>

186. Analysis of environmental data during a lagrangian experiment: The influence of vertical movements

☐ Deep Sea Research II

☐ Abstract:

Publication date: June 2026Source: Deep Sea Research Part II: Topical Studies in Oceanography, Volume 227Author(s): Patricia Romero Fernández, Eulogio Vargas-García, Ricardo Borrego-Santos, Raúl Laiz-Carrión, José Quintanilla, Michael R. Landry, Sven A. Kranz, Michael R. Stukel, Thomas B. Kelly, Manuel Vargas-Yáñez

☐ <https://www.sciencedirect.com/science/article/pii/S0967064526000408>

187. Evaluating the Impacts of Agriculture Conservation on Water Quantity and Quality Through Trend, Predictability, and Causality Analysis

☐ Water Resources Research | Mon, 06 Apr 2026 05:

☐ <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025WR040751>

188. Characterizing Patterns of Drought Synchronicity in the Contiguous United States

☐ Water Resources Research | Mon, 06 Apr 2026 05:

☐ <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025WR041240>

【保护与生物多样性】

189. Conservation Priority Areas Based on Phylogenetic Diversity Are Sensitive to Methods of Phylogenetic Inference and Divergence Time Estimation

☐ Diversity and Distributions | Tue, 31 Mar 2026 00:

☐ <https://onlinelibrary.wiley.com/doi/10.1111/ddi.70166>

190. Functional Traits as Predictors of Global Dominance and Prevalence in Herbaceous Plants

☐ Journal of Biogeography | Sat, 04 Apr 2026 01:

☐ <https://onlinelibrary.wiley.com/doi/10.1111/jbi.70207>

191. Identification of Genetic Variation Associated With Heat Tolerance in Cowpea (*Vigna unguiculata* L. Walp.)

☐ Molecular Ecology | Fri, 03 Apr 2026 04:

☐ <https://onlinelibrary.wiley.com/doi/10.1111/mec.70327>

192. What Drives Conservation Adoption? Social Science Insights from Cattle Ranchers in the Pantanal Wetland, Brazil

☐ Conservation Letters | Fri, 20 Mar 2026 03:

☐ Abstract:

Conservation Letters, Volume 19, Issue 2, March/April 2026.

☐ <https://conbio.onlinelibrary.wiley.com/doi/10.1111/con4.70033>

193. Assessing Spatiotemporal Side Effects of Diversionary Feeding on Mesopredators

☐ Animal Conservation | Sat, 14 Mar 2026 02:

☐ Abstract:

Animal Conservation, EarlyView.

☐ <https://zslpublications.onlinelibrary.wiley.com/doi/10.1111/acv.70059>

【海洋与水生】

194. Relationship between Symbiodiniaceae chlorophyll a concentration and cell density in scleractinian corals across regions, species, and health status

☐ Marine Biology | 2026-04-04

☐ <https://link.springer.com/article/10.1007/s00227-026-04822-9>

【保护与生物多样性】

195. Sleep strengthens successor representations of learned sequences in humans

☐ PLOS Biology | 2026-04-07T14:00:00Z

☐ Abstract:

by Xianhui He, Philipp K. Büchel, Simon Faghel-Soubeyrand, Janina Klingspohr, Marcel S. Kehl, Bernhard P. Staresina Experiences reshape our internal representations of the world. However, the neural and cognitive dynamics of this process are largely unknown. Here, we investigated how sequence learning reorganizes neural representations and how sleep-related consolidation mechanisms contribute to this transformation. Using high-density electroencephalography and multivariate decoding, we found that learning temporal sequences of visual information led to the incorporation of successor representations during a subsequent perceptual task, despite temporal information being task-irrelevant. Importantly, individuals with better sequence memory performance exhibited stronger successor incorporat

☐ <https://journals.plos.org/plosbiology/article>

【植物与农业】

196. An Energy Signalling Pathway Regulates the Aerenchyma Formation in Roots Under Environmental Stresses—A Perspective

☐ Plant Cell & Environment | Mon, 06 Apr 2026 20:

☐ Abstract:

Plant, Cell & Environment, EarlyView.

☐ <https://onlinelibrary.wiley.com/doi/10.1111/pce.70519>

197. The phylogenetic structure of plant communities drives the belowground transmission of fungal pathogens

☐ New Phytologist | Mon, 06 Apr 2026 21:

☐ Abstract:

New Phytologist, EarlyView.

☐ <https://nph.onlinelibrary.wiley.com/doi/10.1111/nph.71156>

【土壤与生物地球化学】

198. Elevated CO₂ does not change the rhizosphere effect on gross nitrogen and phosphorus mobilization in a mature Eucalyptus woodland

☐ Soil Biology and Biochemistry

☐ Abstract:

Publication date: July 2026 Source: Soil Biology and Biochemistry, Volume 218 Author(s): Tobias Rütting, Louise Rütting, Klaus Jarosch, Johanna Pihlblad, Catriona A. Macdonald, C. Florian Stange, Yolima Carrillo

☐ <https://www.sciencedirect.com/science/article/pii/S0038071726000805>

【植物与农业】

199. Dynamic homoeolog expression partitioning contributes to soil-driven adaptive plasticity in allopolyploid *Leymus chinensis*

☐ Plant Journal | Mon, 06 Apr 2026 23:

☐ <https://onlinelibrary.wiley.com/doi/10.1111/tpj.70848>

【土壤与生物地球化学】

200. The effect of modeller's choices and measurement errors on the performance of MIR soil spectroscopy models

☐ Geoderma

☐ Abstract:

Publication date: May 2026 Source: Geoderma, Volume 469 Author(s): Cynthia C.E. van Leeuwen, Vera L. Mulder, Gerard B.M. Heuvelink

☐ <https://www.sciencedirect.com/science/article/pii/S0016706126001321>

【遥感与模型】

201. Annual and diurnal temperature cycle modelling of a merged multi-annual ECOSTRESS and Landsat land surface temperature dataset

☐ International Journal of Applied Earth Observation

☐ Abstract:

Publication date: May 2026 Source: International Journal of Applied Earth Observation and Geoinformation, Volume 149 Author(s): Lluís Pérez-Planells, Frank-M. Göttsche, Jan Cermak

☐ <https://www.sciencedirect.com/science/article/pii/S1569843226001925>

【预印本】

202. Genetic variation in early-season leaf photosynthesis in sugar beet and its relationship with *Cercospora* leaf spot resistance

☐ bioRxiv Plant Biology | 2026-04-06

☐ Abstract:

Improving photosynthesis is a promising approach to enhance sugar beet productivity. However, genetic variation in leaf photosynthesis and its relationship with disease resistance remain underexplored. We evaluated 98 sugar beet genotypes representing different breeding categories, including commercial F1 hybrids, seed-parent lines, and pollinator lines, in Hokkaido, northern Japan. Leaf gas exchange was measured during early growth under field conditions around the infection period of *Cercospora* leaf spot (CLS). To account for fluctuating irradiance during large-scale phenotyping, we applied a multilevel mixed-effects light-response model to estimate genotype-specific photosynthetic characteristics. Substantial genotypic variations in photosynthetic characteristics were detected. F1 hybr

☐ <https://doi.org/10.1101/2026.04.03.716265>

203. Mitigating text data privacy risks from gradient and model inversion attacks with a dual-pronged defense

☐ Research Square - Environment | 2026-04-07 21:03:18

☐ Abstract:

Gradient inversion and model inversion attacks pose serious privacy risks to text data by recovering training samples during or after model training. Existing defenses mainly rely on perturbation or regularization, but most are tailored to a single attack setting, often incur unfavorable privacy-utility trade-offs, and may not directly apply to text data. To address these issues, we propose DGIMI (Defense against Gradient Inversion and Model Inversion), a defense framework that mitigates both attacks on text data while preserving model utility. Our framework is motivated by the fact that input information is encoded in intermediate representations, exposed in gradients during training, and gradually memorized in model parameters. Accordingly, DGIMI intervenes along the propagation of input

☐ <https://www.researchsquare.com/article/rs-9243611/latest>

【综合顶刊】

204. autoFISH: a modular toolbox for sequential single-molecule RNA FISH experiments

☐ Communications Biology | 2026-04-06

☐ Abstract:

Communications Biology, Published online: 06 April 2026; doi:10.1038/s42003-026-09979-zAutoFISH provides a cost-effective, open-source toolbox for automated smFISH, offering integrated hardware, software, and optimized protocols to streamline spatial gene-expression profiling in cell lines and cleared tissue samples.

☐ <https://doi.org/10.1038/s42003-026-09979-z>

205. Separating selection from mutation in antibody language models

☐ eLife | Tue, 07 Apr 2026 00:

☐ Abstract:

Antibodies are encoded by nucleotide sequences that are generated by V(D)J recombination and evolve according to mutation and selection processes. Existing antibody language models, however, focus exclusively on antibodies as strings of amino acids and are fitted using standard language modeling objectives such as masked or autoregressive prediction. In this paper, we first show that fitting models using this objective implicitly incorporates nucleotide-level mutation processes as part of the protein language model, which degrades performance when predicting effects of mutations on functional properties of antibodies. To address this limitation, we devise a new framework: a deep amino acid selection model (DASM) that learns the selection effects of amino acid mutations while explicitly fac

☐ <https://elifesciences.org/articles/109644>

206. Multifaceted bioactivity of brown seaweed-derived fucoidan from the Indian Coastline: A natural health product candidate

☐ PLOS ONE | 2026-04-07T14:00:00Z

☐ Abstract:

by Manisha Prajapati, Dharmendra Kumar, Shivankar Agrawal, Dhiraj Kumar, Pooja Baweja, Dinabandhu Sahoo Brown seaweeds are rich in bioactive constituents, among which fucoidan a sulphated polysaccharide is particularly noted for its broad-spectrum therapeutic properties. This study explores the multifaceted bioactivity of fucoidan extracted from six brown seaweed species (*Sargassum wightii*, *Sargassum tenerrimum*, *Sargassum cinereum*, *Turbinaria conoides*, *Padina boergesenii*, and *Padina tetrastromatica*) collected from the Pudumadam coast, Gulf of Mannar region, Tamil Nadu, India. Fucoidan was isolated and structurally characterized using Fourier-transform infrared spectroscopy (FT-IR) and nuclear magnetic resonance (NMR) spectroscopy. The antioxidant capacity of the extracts was assessed throu

☐ <https://journals.plos.org/plosone/article>

【生态学核心】

207. Dominant and subdominant rooting niches

☐ Journal of Ecology | Mon, 06 Apr 2026 00:

☐ <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2745.70302>

【综合顶刊】

208. Beneficial effects of a novel inspiratory muscle training device on respiratory muscle strength and submaximal functional capacity in community-dwelling older Thai adults: a randomized controlled trial

☐ PeerJ | 2026-04-07

☐ Abstract:

Older adults encounter progressive deterioration of all organ systems. Disease development reportedly increases with age, resulting in decreased physical performance in terms of aerobic endurance, muscular strength, and physical ability. Inspiratory muscle training (IMT) is commonly performed to improve cardiorespiratory fitness. Therefore, this study aimed to develop a resistive inspiratory muscle and determine the effect of inspiratory muscle training on respiratory muscle strength and functional capacity in older adults. Methods Sixty older adults (age: ≥ 60 years), both males and females, were enrolled. All participants performed a respiratory muscle strength test (maximal inspiratory pressure (MIP), maximal expiratory pressure (MEP)) and a 6-minute walking distance test (i.e., submaxim

☐ <https://peerj.com/articles/21072>

【生态学核心】

209. Black juice in the dark: Pollination of dark-nectared *Jasminanthus mucronata* (Apocynaceae) by nocturnal hawkmoths

☐ Ecology | Wed, 01 Apr 2026 05:

☐ <https://esajournals.onlinelibrary.wiley.com/doi/10.1002/ecy.70370>

210. Anthropogenic disturbance modifies spatial patterns of reef sharks

☐ Journal of Applied Ecology | Sun, 05 Apr 2026 21:

☐ <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2664.70330>

211. Consistent Microbial Assembly in the Rhizosphere of Crop Wild Progenitors

☐ Global Ecology and Biogeography | Fri, 27 Mar 2026 21:

☐ <https://onlinelibrary.wiley.com/doi/10.1111/geb.70228>

212. Understanding the effects of patch-burn grazing management on aboveground grassland invertebrate biodiversity

☐ Oikos | Fri, 03 Apr 2026 03:

☐ Abstract:

Oikos, EarlyView.

☐ <https://nsojournals.onlinelibrary.wiley.com/doi/10.1002/oik.11935>

213. Spider mite genotypes with higher growth rate suffer more from competition but exert stronger reproductive interference

☐ Journal of Animal Ecology | Mon, 06 Apr 2026 06:

☐ Abstract:

Journal of Animal Ecology, EarlyView.

☐ <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2656.70247>

214. Development time and host-parasitoid stability: An experimental test

☐ Journal of Animal Ecology | Sun, 05 Apr 2026 22:

☐ Abstract:

Journal of Animal Ecology, EarlyView.

☐ <https://besjournals.onlinelibrary.wiley.com/doi/10.1111/1365-2656.70241>

【环境与气候】

215. Mapping 25 years of global soil salinity: Hydrothermal conditions drive widespread desalinization amid regional salinization hotspots

☐ One Earth

☐ Abstract:

Publication date: Available online 31 March 2026Source: One EarthAuthor(s): Nan Wang, Dietian Yu, Songchao Chen, Jingyi Huang, Ruhollah Taghizadeh-Mehrjardi, Jie Peng, Jean-Pierre Wigneron, Frederic Frappart, Asim Biswas, Zhou Shi

☐ <https://www.sciencedirect.com/science/article/pii/S259033222600059X>

216. Patterns of marine resource conflicts across Africa highlight need for fair access and benefit sharing for a blue economy

☐ One Earth

☐ Abstract:

Publication date: Available online 26 March 2026Source: One EarthAuthor(s): Elizabeth R. Selig, Nahla Gedeon Achi, Frode Sundnes, Colette C.C. Wabnitz, Shinnosuke Nakayama, Dag Ø. Hjermann, Juliano Palacios-Abrantes, Jessica Spijkers, Mafaniso Hara, Moenieba Isaacs, Timothy R. McClanahan, Ethan McKown, Adelina Mensah, Ragnhild Overå, Siri Aas Rustad, Thordis L. Thorarinsdottir, Andreas Forø Tollefsen

☐ <https://www.sciencedirect.com/science/article/pii/S259033222600045X>

217. [ASAP] Hydrodefluorination of a Fluoropolymer by Nonequilibrium Plasma

☐ ES&T Letters | Mon, 30 Mar 2026 03:

☐ Abstract:

Environmental Science & Technology LettersDOI: 10.1021/acs.estlett.6c00275

☐ <https://doi.org/10.1021/acs.estlett.6c00275>

218. [ASAP] Observations of the Gas Phase Composition of the 2024 BioLab Industrial Plume in the Atlanta Metropolitan Area

☐ ES&T Letters | Thu, 26 Mar 2026 02:

☐ Abstract:

Environmental Science & Technology LettersDOI: 10.1021/acs.estlett.6c00160

☐ <https://doi.org/10.1021/acs.estlett.6c00160>

【综合顶刊】

219. Volcano-tectonic earthquake focal mechanisms reveal fluid-induced stress changes driving hydrothermal system development at Mount Ontake

☐ Communications Earth & Environment | 2026-04-07

☐ Abstract:

Communications Earth & Environment, Published online: 07 April 2026; doi:10.1038/s43247-026-03463-6Subsurface fluid movement during the 2024–2025 unrest at Mt. Ontake temporarily reoriented the local stress field beneath the summit, modifying the orientations of fluid-permeable fractures and causing characteristic changes in the focal mechanism patterns of volcano-tectonic earthquakes.

☐ <https://doi.org/10.1038/s43247-026-03463-6>

【环境与气候】

220. Evaluating decoupling Evidence: Examining Timeframes, geographic Scales, and planetary boundaries

☐ Global Environmental Change

☐ Abstract:

Publication date: July 2026Source: Global Environmental Change, Volume 98Author(s): Marina Requena-i-Mora, Dan Brockington

☐ <https://www.sciencedirect.com/science/article/pii/S0959378026000397>

221. Biosourced *Paeonia tenuifolia* L. petal-adsorbent for Pb²⁺ and methyl violet dye removal: kinetical, isothermal and reusability assays

☐ Journal of Environmental Management

☐ Abstract:

Publication date: 15 April 2026 Source: Journal of Environmental Management, Volume 404 Author(s): Aleksandar Jovanović, Jelena Dimitrijević, Natalija Čutović, Janez Kovač, Tatjana Marković, Jelena Petrović

☐ <https://www.sciencedirect.com/science/article/pii/S0301479726010613>

【海洋与水生】

222. The influence of mesoscale oceanographic features on coccolithophore diversity and life cycles in the Southern Indian Ocean

☐ Limnology and Oceanography | Sun, 05 Apr 2026 21:

☐ <https://aslopubs.onlinelibrary.wiley.com/doi/10.1002/lno.70366>

223. The dynamics of microbial communities in the host, water and sediment during the metamorphosis of *Ruditapes philippinarum* larvae

☐ Aquaculture

☐ Abstract:

Publication date: 30 June 2026 Source: Aquaculture, Volume 621 Author(s): Yong Huang, Hongxia Wang, Jun Fang, Pin Huan, Baozhong Liu

☐ <https://www.sciencedirect.com/science/article/pii/S0044848626003479>

224. The unicorn ray: a morphological abnormality in a southern eagle ray (*Myliobatis tenuicaudatus*)

☐ Journal of Fish Biology | Mon, 06 Apr 2026 08:

☐ Abstract:

Journal of Fish Biology, Volume 108, Issue 2, Page 782-789, February 2026.

☐ <https://onlinelibrary.wiley.com/doi/10.1111/jfb.70259>

225. Provenance and geochemistry of quaternary sedimentary components in the Qiongdongnan Basin, South China Sea: discrimination between palaeogeomorphology and sedimentary characteristics

☐ Deep Sea Research I

☐ Abstract:

Publication date: June 2026Source: Deep Sea Research Part I: Oceanographic Research Papers, Volume 229Author(s): Zeyu Lu, Detian Yan

☐ <https://www.sciencedirect.com/science/article/pii/S096706372600035X>

226. Lagrangian perspective on the response of typical chlorophyll structures to mesoscale eddies

☐ Deep Sea Research I

☐ Abstract:

Publication date: June 2026Source: Deep Sea Research Part I: Oceanographic Research Papers, Volume 229Author(s): Yifan Liu, Qiong Xia, Hui Cao, ZeKai Chen

☐ <https://www.sciencedirect.com/science/article/pii/S0967063726000373>

227. Near-inertial wave observations in the Bay of Bengal

☐ Deep Sea Research I

☐ Abstract:

Publication date: June 2026Source: Deep Sea Research Part I: Oceanographic Research Papers, Volume 229Author(s): Hemantha W. Wijesekera, William J. Teague, Zachariah R. Hallock, David W. Wang, Conrad Luecke, Ewa Jarosz, Tommy Jensen, H.J.S. Fernando, S.U.P. Jinadasa

☐ <https://www.sciencedirect.com/science/article/pii/S0967063726000397>

【环境与气候】

228. Perspectives on university-based climate service providers in the United States

☐ Climatic Change | 2026-04-07

☐ <https://link.springer.com/article/10.1007/s10584-026-04150-3>

229. Estimating the impacts of recurrent and expanding coastal flooding on septic systems in Maryland's Chesapeake Bay

☐ Climatic Change | 2026-04-07

☐ <https://link.springer.com/article/10.1007/s10584-026-04172-x>

230. An assessment of climate risks and vulnerabilities of unpaved secondary roads and resilient enhancement strategies in Sierra Leone

☐ Climatic Change | 2026-04-06

☐ <https://link.springer.com/article/10.1007/s10584-026-04149-w>

231. Sea surface salinity variability and its underlying processes in the eastern South Indian Ocean

☐ Climate Dynamics | 2026-03-26

☐ <https://link.springer.com/article/10.1007/s00382-026-08140-0>

232. Mechanisms and seasonal predictability of sea surface temperature in the Senegalo-Mauritanian Upwelling System

☐ Climate Dynamics | 2026-03-24

☐ <https://link.springer.com/article/10.1007/s00382-026-08121-3>

【海洋与水生】

233. EPS-mediated mineralization drives granule densification and enhances denitratation – anammox coupling under alkaline conditions

☐ Water Research

☐ Abstract:

Publication date: 1 July 2026 Source: Water Research, Volume 299 Author(s): Shenbin Cao, Xingping Liu, Yucheng Tao, Jiawei Ren, Zhiwei Zhou, Rui Du

☐ <https://www.sciencedirect.com/science/article/pii/S0043135426005701>

234. Adaptive reinforcement learning for energy-efficient high-recovery closed-circuit reverse osmosis

☐ Water Research

☐ Abstract:

Publication date: 1 July 2026 Source: Water Research, Volume 299 Author(s): Jeongwoo Moon, Byeongchan Yun, Kiho Park, Seong-Su Kim, Youngjoo Lee, Kwanho Jeong, Kyung Hwa Cho

☐ <https://www.sciencedirect.com/science/article/pii/S0043135426005373>

235. Mechanistic insights into sulfur-driven denitrifying phosphorus removal: Roles of zero-valent sulfur and Thiothrix in the DS-EBPR system

☐ Water Research

☐ Abstract:

Publication date: 1 July 2026 Source: Water Research, Volume 299 Author(s): Haoting Quan, Yanjuan Luo, Yanyan Jia, Yifan Zhang, Peng Qi, Yongsun Shi, Lianpeng Sun, Hui Lu

☐ <https://www.sciencedirect.com/science/article/pii/S0043135426005531>

236. Exploring the Transferability of Image-Based Algorithms for River Plastic Detection: The Value of Small Mixed Data Sets

☐ Water Resources Research | Mon, 06 Apr 2026 05:

☐ <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025WR040605>

【保护与生物多样性】

237. Oyster Size Distributions Vary Across Time and Space in the Tampa Bay Estuary, 250–2024 ce

☐ Journal of Biogeography | Sat, 04 Apr 2026 01:

☐ <https://onlinelibrary.wiley.com/doi/10.1111/jbi.70208>

238. Connecting the Dots: Does COI Variability in *Paramecium caudatum* (Ciliates, Protista) Support the Moderate Endemicity Model of Microeukaryote Distribution?

☐ Journal of Biogeography | Sat, 04 Apr 2026 01:

☐ <https://onlinelibrary.wiley.com/doi/10.1111/jbi.70206>

239. Chromosomal Fusions Promote Speciation in Subterranean Blind Mole Rats

☐ Molecular Ecology | Mon, 06 Apr 2026 01:

☐ <https://onlinelibrary.wiley.com/doi/10.1111/mec.70331>

240. Intense Hunting Pressure Drives Roost Use in the Egyptian Fruit Bat: Conservation and One Health Implications

☐ Animal Conservation | Thu, 12 Mar 2026 06:

☐ Abstract:

Animal Conservation, EarlyView.

☐ <https://zslpublications.onlinelibrary.wiley.com/doi/10.1111/acv.70056>

【海洋与水生】

241. Sex-specific predation of the non-native invasive American blue crab *Callinectes sapidus* on the Mediterranean mussel *Mytilus galloprovincialis*

☐ Marine Biology | 2026-03-24

☐ <https://link.springer.com/article/10.1007/s00227-026-04812-x>

【保护与生物多样性】

242. The m6Am methyltransferase PCIF1 promotes osteogenic differentiation of mesenchymal stem cells through stabilization of Wnt-related transcripts

☐ PLOS Biology | 2026-04-06T14:00:00Z

☐ Abstract:

by Wei Song, Kuan-Jui Su, Zhehui Pan, Zhong Zhang, Qi Yin, Weimin Lin, Linfeng Liu, Yun Gong, Bocheng Liang, Yimeng Cai, Qiwen Li, Hui Shen, Hong-Wen Deng, Quan Yuan Osteogenesis depends on the self-renewal and differentiation of mesenchymal stem cells (MSCs). Emerging research underscores the regulatory functions of RNA methylation on bone homeostasis. Here, we show PCIF1, the N6,2'-O-dimethyladenosine (m6Am) methyltransferase, is essential for maintaining bone mass and promoting osteogenic differentiation of MSCs. Multiple complementary analyses—including GWAS, TWAS, and single-cell transcriptomics—collectively point to PCIF1 as a regulator of human bone mineral traits and early-stage mesenchymal differentiation. Global or MSC-specific *Pcif1* deletion elicits osteoporotic pathology in mic

☐ <https://journals.plos.org/plosbiology/article>

【植物与农业】

243. Short-term legume cover crops following maize only marginally impact winter cereals under conservation tillage management

☐ Field Crops Research

☐ Abstract:

Publication date: 15 May 2026Source: Field Crops Research, Volume 342Author(s): Arun D. Jani, John E. Erickson, Ramon G. Leon, C. Wesley Wood, Diane Rowland, Heather A. Enloe, Michael J. Mulvaney

☐ <https://www.sciencedirect.com/science/article/pii/S0378429026001474>

244. Membrane interaction of cyanobacterial and chloroplast ESCRT-III proteins

☐ Plant Journal | Mon, 06 Apr 2026 01:

☐ <https://onlinelibrary.wiley.com/doi/10.1111/tpj.70843>

245. Activation of ethylene biosynthetic gene MaACS1 by RING-H2-type E3 ligase MaRHA2A1-mediated ubiquitin-proteasomal degradation of MaMYB73 during banana fruit ripening

☐ Plant Journal | Mon, 06 Apr 2026 00:

☐ <https://onlinelibrary.wiley.com/doi/10.1111/tpj.70851>

【土壤与生物地球化学】

246. Ecological Variability and Controls on Denitrifying Anaerobic Methane Oxidation

☐ Global Biogeochemical Cycles | Sun, 05 Apr 2026 22:

☐ <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025GB008766>

247. Drivers of Benthic Iron Fluxes and the Pore-Water Iron Isotopic Signature in Surface Sediments of South Georgia Fjords, Sub-Antarctica

☐ Global Biogeochemical Cycles | Sat, 04 Apr 2026 04:

☐ <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025GB008856>

248. Rheological evidence of wetting-drying-driven microstructural reorganization in Oxisols: The role of iron oxide crystallinity and structural integrity

☐ Soil & Tillage Research

☐ Abstract:

Publication date: October 2026Source: Soil and Tillage Research, Volume 262Author(s): Raí Ferreira Batista, Alan Carlos Batistão, Amanda Romeiro Alves, Fabrício de Araújo Pedron, José Miguel Reichert

☐ <https://www.sciencedirect.com/science/article/pii/S0167198726001492>

249. Herbicide applications in broccoli (*Brassica oleracea* var. *italica*) using Unmanned Aerial Systems

☐ Soil & Tillage Research

☐ Abstract:

Publication date: September 2026Source: Soil and Tillage Research, Volume 261Author(s): Fatemeh Esmaeil-beiki, Vijay Singh, Akashdeep Brar, Daniel E. Martin, Mark Reiter

☐ <https://www.sciencedirect.com/science/article/pii/S0167198726001467>

250. Grassland gullies develop faster and cause more severe land destruction than cropland gullies in the agropastoral ecotone of northeastern China

☐ Soil & Tillage Research

☐ Abstract:

Publication date: September 2026Source: Soil and Tillage Research, Volume 261Author(s): Jie Tang, Hanquan Cheng, Yun Xie, Wengting Wang, Shaoqing Liu, Wei Ouyang, Yan Gao

☐ <https://www.sciencedirect.com/science/article/pii/S0167198726001479>

【遥感与模型】

251. Studying domain shifts in bioacoustic recording style between focal and soundscape recordings with acoustic indices

☐ Ecological Informatics

☐ Abstract:

Publication date: May 2026 Source: Ecological Informatics, Volume 95 Author(s): Sean Perry, Tianqi Zhang, Siya Kamboj, Anu Jajodia, Dhruv Tomar, Ryan Kastner

☐ <https://www.sciencedirect.com/science/article/pii/S1574954126001457>

252. Blocked Coastal Propagation Inhibits Model Representation of Southeast United States Coastal Sea Level Variability

☐ Geophysical Research Letters | Mon, 06 Apr 2026 05:

☐ Abstract:

Geophysical Research Letters, Volume 53, Issue 7, 16 April 2026.

☐ <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025GL118781>

【预印本】

253. A pseudo-phased genome assembly for *Hemileia vastatrix* reveals an isolate-specific chromosomal haploid trisomy

☐ bioRxiv Genomics | 2026-04-07

☐ Abstract:

Recurrent epidemics of coffee leaf rust, caused by the fungal pathogen *Hemileia vastatrix*, have constrained production of Arabica coffee for over 150 years. Here, we present a pseudo-phased, chromosome-level genome resource for *H. vastatrix*, isolate Hv178a, to guide research into disease management. The Hv178a genome assembly is 665 and 638 Mbp for haplotype A and B respectively, localised to 18 chromosomes. We determined that the genomes are highly repetitive at ~90%, with a GC content of ~33%. We present the full annotation of 13,760 and 17,998 protein coding genes, and we predicted 452 and 496 effectors in haplotype A and B respectively. Depth-based comparisons with 11 additional *H. vastatrix* isolates revealed increased chromosome 17 (chr17) copy number in Hv178a. Validation with qPCR

☐ <https://doi.org/10.1101/2026.04.04.716458>

254. Barriers and Facilitators to Choosing Traditional Chinese Medicine among Patients with Degenerative Lumbar Spinal Stenosis in China: A Qualitative Study

☐ Research Square - Environment | 2026-04-07 21:44:30

☐ Abstract:

Purpose This study aimed to qualitatively explore the perceived barriers and facilitators that influence decision-making among patients with degenerative lumbar spinal stenosis (DLSS) in China regarding the use of Traditional Chinese Medicine (TCM) interventions. **Methods** A qualitative descriptive study was conducted using semi-structured interviews. Through purposive sampling, 26 patients with DLSS were recruited from a tertiary TCM hospital in Beijing. Interviews explored patients' experiences, perceptions, and the reasoning behind their decisions to use or decline TCM interventions (e.g., acupuncture, tuina). Data were analyzed using inductive thematic analysis. **Results** Patient decision-making emerged as a dynamic process of weighing interlinked factors. Barriers included: (1) skepticism a

☐ <https://www.researchsquare.com/article/rs-8812194/latest>

【生态学核心】

255. Biogeography of intertidal invertebrates is influenced by latitude along the west coast of Australia

☐ Ecography | Fri, 03 Apr 2026 03:

☐ Abstract:

Ecography, EarlyView.

☐ <https://nsojournals.onlinelibrary.wiley.com/doi/10.1002/ecog.08383>

【环境与气候】

256. Unfinished buildings as a growing challenge for urban planetary health

☐ One Earth

☐ Abstract:

Publication date: Available online 24 March 2026Source: One EarthAuthor(s): Beijia Huang, Yuqiong Long, Zhi Cao

☐ <https://www.sciencedirect.com/science/article/pii/S2590332226000576>

【海洋与水生】

257. Dietary hesperetin mitigates high-fat-induced growth suppression, intestinal lipid accumulation, oxidative stress, and inflammation in largemouth bass (*Micropterus salmoides*)

☐ Aquaculture

☐ Abstract:

Publication date: 30 June 2026Source: Aquaculture, Volume 621Author(s): Peng Zhao, Chong-Chao Zhong, Nermeen Abu-Elala, Yu-Chen Zhou, Zhi-Wei Hao, Zhi Luo, Xiao-Ying Tan

☐ <https://www.sciencedirect.com/science/article/pii/S0044848626003492>

258. Feeding as an alternative to injection for RNAi in mussels: A comparative evaluation of gene-silencing efficacy and integrated physiological impacts

☐ Aquaculture

☐ Abstract:

Publication date: 30 June 2026Source: Aquaculture, Volume 621Author(s): Ruoxuan Zhao, Qing Fang, Shuimei Li, Kaijie Wu, Liya Li, Yanan Di

☐ <https://www.sciencedirect.com/science/article/pii/S0044848626003467>

【环境与气候】

259. Interdecadal Pacific Oscillation modulates the interdecadal variability of the Arctic sea ice dipole

☐ Climate Dynamics | 2026-03-26

☐ <https://link.springer.com/article/10.1007/s00382-026-08132-0>

【海洋与水生】

260. What is the contribution of thermal selection to the maintenance of a shell color cline in *Littorina saxatilis*?

☐ Marine Biology | 2026-03-25

☐ <https://link.springer.com/article/10.1007/s00227-026-04808-7>

【保护与生物多样性】

261. The time has come for big changes to improve research funding

☐ PLOS Biology | 2026-04-06T14:00:00Z

☐ Abstract:

by Peter Kolarz The competitive research funding system is at a breaking point. Innovations to address ongoing problems are needed on a grander scale than ever before, but even this will not suffice to fix a struggling system. What we need is a whole system transformation. The competitive research funding system is at breaking point. This Perspective argues that innovations to address ongoing problems are needed on a grander scale than ever before, but this will not suffice. What we need is whole system transformation.

☐ <https://journals.plos.org/plosbiology/article>

【植物与农业】

262. Surrogate to Poincaré inequalities on manifolds for dimension reduction in nonlinear feature spaces

☐ Plant Cell | Mon, 16 Mar 2026 00:

☐ Abstract:

Abstract We aim to approximate a continuously differentiable function $u: \mathbb{R}^d \rightarrow \mathbb{R}$ by a composition of functions $f \circ g$ where $g: \mathbb{R}^d \rightarrow \mathbb{R}^m$, $m \leq d$ and $f: \mathbb{R}^m \rightarrow \mathbb{R}$ are built in a two stage procedure. For a fixed g , we build f using classical regression methods, involving evaluations of u . Recent works proposed to build a nonlinear g by minimizing a loss function $\mathcal{J}(g)$ derived from Poincaré inequalities on manifolds, involving evaluations of the gradient of u . A problem is that minimizing $\mathcal{J}$ may be a challenging task. Hence in this work, we introduce new convex surrogates to $\mathcal{J}$. Leveraging concentration inequalities, we provide suboptimality results for

☐ <https://academic.oup.com/imaiai/article/doi/10.1093/imaiai/iaag005/8524968>

263. Gaussian random projections of convex cones: approximate kinematic formulae and applications

☐ Plant Cell | Mon, 16 Mar 2026 00:

☐ Abstract:

Abstract Understanding the stochastic behavior of random projections of geometric sets constitutes a fundamental problem in high dimension probability that finds wide applications in diverse fields. This paper provides a kinematic description for the behavior of Gaussian random projections of closed convex cones, in analogy to that of randomly rotated cones studied in Amelunxen et al. (2014, Inf. Inference J. IMA, 3, 224 – 294). Formally, let K be a closed convex cone in $\mathbb{R}^n$, and $G \in \mathbb{R}^{m \times n}$ be a Gaussian matrix with i.i.d. $\mathcal{N}(0, 1)$ entries. We show that $GK \equiv G \mu : \mu \in K$ behaves like a randomly rotated cone in $\mathbb{R}^m$ with statistical dimension $\min\{\delta(K), m\}$, in the following kinematic sense: for any fixed closed cone

☐ <https://academic.oup.com/imaiai/article/doi/10.1093/imaiai/iaag006/8524962>

264. Empirical Bernstein and betting confidence intervals for randomized quasi-Monte Carlo

☐ Plant Cell | Fri, 06 Mar 2026 00:

☐ Abstract:

Abstract Randomized quasi-Monte Carlo (RQMC) methods estimate the mean of a random variable by sampling an integrand at n equidistributed points. For scrambled digital nets, the resulting variance is typically $\tilde{O}(n^{-\theta})$, where $\theta \in [1, 3]$ depends on the smoothness of the integrand and $\tilde{O}$ neglects logarithmic factors. While RQMC can be far more accurate than plain Monte Carlo (MC), it remains difficult to get confidence intervals on RQMC estimates. We investigate some empirical Bernstein confidence intervals (EBCIs) and hedged betting confidence intervals (HBCIs), both from Waudby-Smith and Ramdas (2024, J. Roy. Statist. Soc. B, 86, 1–27), when the random variable of interest is subject to known bounds. When there are N integrand evaluations partitioned into

☐ <https://academic.oup.com/imaiai/article/doi/10.1093/imaiai/iaag003/8509318>

265. DRB4-dependent trans-acting siRNAs regulate lemma polarity development in rice

☐ New Phytologist | Mon, 06 Apr 2026 22:

☐ Abstract:

New Phytologist, EarlyView.

☐ <https://nph.onlinelibrary.wiley.com/doi/10.1111/nph.71127>

【预印本】

266. Septins regulate cytokinesis and multicellular development in the closest living relatives of animals

☐ bioRxiv Evolutionary Biology | 2026-04-07

☐ Abstract:

Septins are cytoskeletal proteins that regulate cytokinesis in fungi and animals, yet their functions in choanoflagellates - the closest living relatives of animals - have remained unknown. *Salpingoeca rosetta*, a choanoflagellate that switches between unicellular and multicellular forms, encodes four septins closely related to animal and fungal septins. CRISPR/Cas9-mediated disruption of *S. rosetta* septins revealed that a subset regulate cell size, with two mutants exhibiting an elevated frequency of oversized cells and one exhibiting smaller cells. Three of the four septins were required for proper rosette colony development, while two also regulated rosette structural integrity. Characterization of *Sros_septA*, which showed the strongest phenotype, revealed a role in cytokinesis: mutant c

☐ <https://doi.org/10.1101/2026.04.05.716596>

【遥感与模型】

267. A Physics-Informed Neural Network Approach to the Gannon Storm

☐ Geophysical Research Letters | Mon, 06 Apr 2026 18:

☐ Abstract:

Geophysical Research Letters, Volume 53, Issue 7, 16 April 2026.

☐ <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025GL121605>

268. Spatiotemporal Variations in the Interplate Slip Rate Around Kodiak Island, Alaska

☐ Geophysical Research Letters | Mon, 06 Apr 2026 04:

☐ Abstract:

Geophysical Research Letters, Volume 53, Issue 7, 16 April 2026.

☐ <https://agupubs.onlinelibrary.wiley.com/doi/10.1029/2025GL120066>

【预印本】

269. Against Extraction: Indigenous Modernism in the Twin Cities

☐ EarthArXiv | Thu, 02 Apr 2026 08:

☐ <https://journalpub.escholarship.org/aicrj/article/id/64276/>

270. By the Fire We Carry: The Generations-Long Fight for Justice on Native Land

☐ EarthArXiv | Thu, 02 Apr 2026 08:

☐ <https://journalpub.escholarship.org/aicrj/article/id/63648/>

271. Prison Born: Incarceration and Motherhood in the Colonial Shadow

☐ EarthArXiv | Thu, 02 Apr 2026 08:

☐ <https://journalpub.escholarship.org/aicrj/article/id/62969/>

272. Hematocrit as the Best Predictor of Red Blood Cell Transfusion in Elderly Patients with Femoral Neck Fractures Undergoing Hip Hemiarthroplasty

☐ Research Square - Environment | 2026-04-07 21:34:26

☐ Abstract:

Background Overestimation of blood transfusion in older patients with femoral neck fractures undergoing hip hemiarthroplasty results in unnecessary consumption of blood bank resources. Herein, we aimed to develop a preoperative predictive tool for appropriate red blood cell preparation. Methods A retrospective cohort study was conducted between January 2015 to October 2025. Univariate and multivariate logistic regression were used to identify predictive factors. The area under the receiver operating characteristic curve (AuROC) was used to evaluate significant factors. Predictive performance was validated using sensitivity, specificity, and positive and negative predictive

values, and externally validated in independent cases. Results Among 424 patients, 135 patients received blood transfusi

☐ <https://www.researchsquare.com/article/rs-9144493/latest>

273. Against Extraction: Indigenous Modernism in the Twin Cities

☐ EcoEvoRxiv | Thu, 02 Apr 2026 08:

☐ <https://journalpub.escholarship.org/aicrj/article/id/64276/>

274. By the Fire We Carry: The Generations-Long Fight for Justice on Native Land

☐ EcoEvoRxiv | Thu, 02 Apr 2026 08:

☐ <https://journalpub.escholarship.org/aicrj/article/id/63648/>

275. Prison Born: Incarceration and Motherhood in the Colonial Shadow

☐ EcoEvoRxiv | Thu, 02 Apr 2026 08:

☐ <https://journalpub.escholarship.org/aicrj/article/id/62969/>